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FRACTURE PLATING SYSTEMS AND METHODS

Abstract

A fracture plating system may be configured to stabilize a first fracture of a bone of a patient, the bone may include an interior surface facing toward an interior body cavity of the patient, and an exterior surface facing away from the interior body cavity. The fracture plating system may include a plate having one or more slots and configured to span the first fracture, a first fastener configured to be received in one of the one or more slots and secure the plate to the interior surface, a second fastener configured to be received in one of the one or more slots and secure the plate to the interior surface, and a first tether configured to apply a force directly to the first and second fasteners to guide the plate, the first and second fasteners through the interior body cavity to the interior surface of the bone.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application claims the benefit of U.S. Provisional Application No. 63/560,219 filed on Mar. 1, 2024, entitled “FRACTURE PLATING SYSTEMS AND METHODS”; U.S. Provisional Application No. 63/572,491 filed on Apr. 1, 2024, entitled “FRACTURE PLATING SYSTEMS AND METHODS”; U.S. Provisional Application No. 63/693,769 filed on Sep. 12, 2024, entitled “INTERNAL RIB PLATING SYSTEMS AND METHODS”; and U.S. Provisional Application No. 63/748,317 filed on Jan. 22, 2025, entitled “RIB FIXATION SYSTEMS AND METHODS”. The foregoing documents are hereby incorporated by reference in their entirety.

TECHNICAL FIELD
[0002] The present disclosure relates generally to surgical systems and methods, and more particularly, to systems and methods for fracture plating and stabilization.
BACKGROUND
[0003] Bone fractures, particularly in anatomical regions such as the ribs, can present significant challenges in treatment due to their anatomical complexity, constant motion during respiration, and the difficulty of achieving stable fixation. Rib fractures, flail chest, and other thoracic injuries often result from trauma and can lead to severe pain, respiratory complications, and impaired pulmonary function. Proper stabilization of fractured rib segments is crucial to facilitate healing, reduce pain, and restore respiratory mechanics.
[0004] Conventional methods of treating rib fractures often involve conservative management, including pain control and respiratory therapy. However, in cases of severe fractures, such as flail chest or displaced rib fractures, surgical intervention may be necessary. Traditional open surgical approaches for rib fixation typically require large incisions, displacement of muscle tissue, and exposure of the fracture site, which can result in increased morbidity, prolonged recovery times, and additional complications such as infection and tissue damage.
[0005] Percutaneous fixation techniques have emerged as a less invasive alternative, offering potential advantages such as reduced surgical trauma, shorter recovery times, and lower complication rates. However, existing percutaneous bone fixation systems are often limited in their ability to securely stabilize bone segments, particularly in regions like the ribs where anatomical curvature and constant mechanical forces present additional challenges. Many current devices and methods lack the necessary adaptability and stability to accommodate complex fracture patterns and ensure proper alignment and fixation of bone segments. There is a need for an improved bone repair system designed specifically for percutaneous fixation of bone segments, such as rib bones.
SUMMARY
[0006] The various systems and methods of the present disclosure have been developed in response to the present state of the art, and in particular, in response to the problems and needs in the art that have not yet been fully solved by currently available fracture plating systems and methods.
[0007] In some embodiments, a fracture plating system may be configured to stabilize a first fracture of a bone of a patient, the bone may include an interior surface facing toward an interior body cavity of the patient, and an exterior surface facing away from the interior body cavity. The fracture plating system may include a plate having one or more slots and configured to span the

first fracture, a first fastener configured to be received in one of the one or more slots and secure the plate to the interior surface of the bone, a second fastener configured to be received in one of the one or more slots and secure the plate to the interior surface of the bone, and a first tether configured to apply a force directly to the first fastener and the second fastener to guide the plate, the first fastener, and the second fastener through the interior body cavity to the interior surface of the bone.

[0008] In the fracture plating system of any preceding paragraph, the first tether may be further configured to facilitate reduction of the first fracture by, with the first fastener and the second fastener secured to the interior surface on opposite sides of the first fracture, urging the first fastener and the second fastener towards each other.

[0009] In the fracture plating system of any preceding paragraph, the first tether may be further configured to guide the first fastener through a first hole in the bone and to guide the second fastener through a second hole in the bone, wherein the first hole and the second hole may be on opposite sides of the first fracture.

[0010] In the fracture plating system of any preceding paragraph, the bone may further include a second fracture. The plate may be further configured to stabilize the second fracture via attachment of the plate to the bone such that the plate may span the first fracture and the second fracture, and the fracture plating system may further include a third fastener configured to be received in one of the one or more slots and secure the plate to the interior surface of the bone, a fourth fastener configured to be received in one of the one or more slots and secure the plate to the interior surface of the bone; and a second tether configured to guide the third fastener and the fourth fastener to the interior surface of the bone.

[0011] In the fracture plating system of any preceding paragraph, the second tether may be further configured to guide the third fastener to a first slot of the one or more slots and to guide the fourth fastener to a second slot of the one or more slots.

[0012] In the fracture plating system of any preceding paragraph, the fracture plating system may further be configured to stabilize three or more fractures of the bone, wherein the plate may be further configured to span the three or more fractures and the fracture plating system may further include multiple tethers each configured to guide two fasteners to the interior surface of the bone and to one of the one or more slots.

[0013] In the fracture plating system of any preceding paragraph, the bone may include a rib, and the first tether may be further configured to draw the plate, the first fastener, and the second fastener through a VATS portal to the interior surface of the bone.

[0014] In some embodiments, a fracture plating system may be configured to stabilize a first fracture of a bone of a patient, the bone may include an interior surface facing toward an interior body cavity of the patient, and an exterior surface facing away from the interior body cavity. The fracture plating system may include a plate having one or more slots and configured to span the first fracture, a first fastener configured to be received in one of the one or more slots and secure the plate to the interior surface of the bone, and a first tether configured to apply a force directly to the first fastener and the second fastener to guide the plate, the first fastener, and the second fastener through the interior body cavity to the interior surface of the bone.

[0015] In the fracture plating system of any preceding paragraph, the first tether may be further configured to facilitate reduction of the first fracture by, with the first fastener and the second fastener secured to the interior surface on opposite sides of the first fracture, urging the first fastener and the second fastener towards each other.

[0016] In the fracture plating system of any preceding paragraph, the bone may include a rib, and the first tether may be further configured to draw the plate, the first fastener, and the second fastener through a VATS portal to the interior surface of the bone.

[0017] In the fracture plating system of any preceding paragraph, the bone may further include a second fracture, and the plate may be further configured to stabilize the second fracture via attachment of the plate to the bone such that the plate spans the first fracture and the second fracture. The fracture plating system may further include a third fastener configured to be received in one of the one or more slots and secure the plate to the interior surface of the bone, a fourth fastener configured to be received in one of the one or more slots and secure the plate to the interior surface of the bone, and a second tether configured to guide the third fastener and the fourth fastener to the interior surface of the bone.

[0018] In the fracture plating system of any preceding paragraph, the first fastener may include a first cannulation configured to receive a first end of the first tether, and the second fastener may include a second cannulation configured to receive a second end of the first tether.

[0019] In the fracture plating system of any preceding paragraph, the fracture plating system may further include a first locking nut configured to receive the first fastener and to cooperate with the first fastener to secure the plate to the interior surface of the bone, and a second locking nut configured to receive the second fastener and to cooperate with the second fastener to secure the plate to the interior surface of the bone.

[0020] In some embodiments, a fracture plating system may be configured to stabilize a first fracture of a bone of a patient, the bone may include an interior surface facing toward an interior body cavity of the patient, and an exterior surface facing away from the interior body cavity. The fracture plating system may include a plate assembly having a plate including one or more slots and configured to span the first fracture, a first fastener coupled to one of the one or more slots and configured to secure the plate to the interior surface of the bone, and a second fastener coupled to one of the one or more slots and configured to secure the plate to the interior surface of the bone. The fracture plating system may also include a first tether configured to apply a force directly to the first fastener and the second fastener to guide the plate assembly through the interior body cavity to the interior surface of the bone.

[0021] In the fracture plating system of any preceding paragraph, the first tether may be further configured to draw the first fastener through a first hole in a first portion of the bone proximate a first side of the first fracture; and draw the second fastener through a second hole in a second portion of the bone proximate a second side of the first fracture.

[0022] In the fracture plating system of any preceding paragraph, the bone may further include a second fracture, and the plate may be further configured to stabilize the second fracture via attachment of the plate to the bone such that the plate spans the first fracture and the second fracture. The fracture plating system may further include a third fastener configured to be received in one of the one or more slots and secure the plate to the interior surface of the bone, a fourth fastener configured to be received in one of the one or more slots and secure the plate to the interior surface of the bone, and a second tether configured to guide the third fastener and the fourth fastener to the interior surface of the bone.

[0023] In the fracture plating system of any preceding paragraph, the first tether may be further configured to facilitate reduction of the first fracture by, with the first fastener and the second fastener secured to the interior surface on opposite sides of the first fracture, urging the first fastener and the second fastener towards each other.

[0024] In the fracture plating system of any preceding paragraph, the bone may include a rib, and the first tether may be further configured to draw the plate assembly through a VATS portal to the interior surface of the bone.

[0025] In the fracture plating system of any preceding paragraph, the first fastener may include a first cannulation configured to receive a first end of the first tether, and the second fastener may include a second cannulation configured to receive a second end of the first tether.

[0026] In the fracture plating system of any preceding paragraph, the fracture plating system may further include a first locking nut configured to receive the first fastener and to cooperate with the first fastener to secure the plate to the interior surface of the bone, and a second locking nut configured to receive the second fastener and to cooperate with the second fastener to secure the plate to the interior surface of the bone.

[0027] In some embodiments, a fracture plating system may be configured to stabilize a first fracture of a bone of a patient, the bone may include an interior surface facing toward an interior body cavity of the patient, and an exterior surface facing away from the interior body cavity. The fracture plating system may include a first plate having a first slot having a first slot width, a first slot length, a first longitudinal axis, and a first threaded portion, wherein the first plate may be configured to span the first fracture, and a first fastener having a first head portion and a second threaded portion, wherein the first fastener may be configured to be received in the first slot and secure the first plate to the interior surface of the bone. The first slot width and the first head portion may be sized to prevent the first head portion from passing through the first slot, and the first threaded portion may be configured to threadably engage the second threaded portion to allow the second threaded portion to threadably pass through the first threaded portion, thereby positioning the first slot between the first head portion and the second threaded portion, resulting in the first fastener being captive in the first slot.

[0028] In the fracture plating system of any preceding paragraph, the first fastener and the first slot may be configured to allow translation of the first fastener along the first longitudinal axis, with the first fastener captively received in the first slot, and the first fastener may be further configured so that, with the first fastener captively received in the first slot, the first fastener may be moveable relative to the first plate along a third longitudinal axis of the first fastener.

[0029] In the fracture plating system of any preceding paragraph, the first fastener may be configured to reside captive within the first slot independently of engagement of any protruding feature of the first head portion with the first plate.

[0030] In the fracture plating system of any preceding paragraph, the fracture plating system may further include a second plate having a second plate length and a second slot, wherein the second plate may be configured to span the first fracture, a second fastener configured to be captively received in the first slot to secure the first plate to the interior surface of the bone or captively received in the second slot to secure the second plate to the interior surface of the bone. One of the first plate and the second plate may be selected based on patient anatomy and a location of the first fracture, two of the first fastener and the second fastener may be selected based on patient anatomy and the location of the first fracture, and the fracture plating system may be configured to be assembled during a surgical procedure prior to implantation within the patient.

[0031] In the fracture plating system of any preceding paragraph, the bone may include a rib, and the fracture plating system may further include a tether configured to draw the first plate and two of the first fastener and the second fastener through a VATS portal to the interior surface of the bone.

[0032] In the fracture plating system of any preceding paragraph, the first fastener may include a first fastener length, and the second fastener may include a second fastener length, different than the first fastener length.

[0033] In the fracture plating system of any preceding paragraph, the fracture plating system may further include a first locking nut having a first body length, and a second locking nut having a second body length, different than the first body length, wherein the first locking nut and the second locking nut may be interchangeably securable to the first fastener and the second fastener.

[0034] In some embodiments, a fracture plating system may be configured to stabilize a first fracture of a bone of a patient, the bone may include an interior surface facing toward an interior body cavity of the patient, and an exterior surface facing away from the interior body cavity. The fracture plating system may include a first plate having a first slot extending along a first longitudinal axis of the first plate, wherein the first plate may be configured to span the first fracture, a first fastener configured to be captively received in the first slot and secure the first plate to the interior surface of the bone, and a first tether configured to guide the first plate and the first fastener to the interior surface of the bone. The first fastener may be configured so that, with the first fastener captively received in the first slot, the first fastener may be moveable relative to the first plate along a second longitudinal axis of the first fastener.

[0035] In the fracture plating system of any preceding paragraph, the first slot may include a first threaded portion and the first fastener may include a second threaded portion and a first head portion, wherein the first threaded portion may be configured to threadably engage the second threaded portion to allow the second threaded portion to pass through the first threaded portion, thereby positioning the first slot between the first head portion and the second threaded portion, resulting in the first fastener being captively received in the first slot.

[0036] In the fracture plating system of any preceding paragraph, the first fastener may include a first head portion, and the first fastener may be configured to reside captive within the first slot independently of engagement of any protruding feature of the first head portion with the first plate.

[0037] In the fracture plating system of any preceding paragraph,, the bone may include a rib, and the first tether may be further configured to draw the first plate and the first fastener through a VATS portal to the interior surface of the bone.

[0038] In the fracture plating system of any preceding paragraph, the fracture plating system may further include a second plate having a second plate length and a second slot extending along a third longitudinal axis of the second plate, wherein the second plate may be configured to span the first fracture, a second fastener having a second fastener length and configured to be captively received in the first slot to secure the first plate to the interior surface of the bone or captively received in the second slot to secure the second plate to the interior surface of the bone. The first fastener may include a first fastener length, different than the second fastener length, one of the first plate and the second plate may be selected based on patient anatomy and a location of the first fracture, two of the first fastener and the second fastener may be selected based on patient anatomy and the location of the first fracture, and the fracture plating system may be configured to be assembled during a surgical procedure prior to implantation within the patient.

[0039] In the fracture plating system of any preceding paragraph, the fracture plating system may further include a first locking nut having a first body length, and a second locking nut having a second body length, different than the first body length, wherein the first locking nut and the second locking nut may be interchangeably securable to the first fastener and the second fastener.

[0040] In some embodiments, a fracture plating system may be configured to stabilize a first fracture of a bone of a patient, the bone may include an interior surface facing toward an interior body cavity of the patient, and an exterior surface facing away from the interior body cavity. The fracture plating system may include a first plate having a first plate length and a first slot extending along a first longitudinal axis of the first plate, wherein the first plate may be configured to span the first fracture, a second plate having a second plate length, different from the first plate length, and a second slot extending along a second longitudinal axis of the second plate, wherein the second plate may be configured to span the first fracture, a first fastener configured to be captively receivable in either of the first slot and the second slot to secure either of the first plate and the second plate to the interior surface of the bone, and a second fastener configured to be captively receivable in either of the first slot and the second slot to secure either of the first plate and the second plate to the interior surface of the bone.

[0041] In the fracture plating system of any preceding paragraph, the first slot may include a first threaded portion and a first slot width, and the first fastener may include a second threaded portion and a head portion. The first slot width and the head portion may be sized to prevent the head portion from passing through the first slot, and the first threaded portion may be configured to threadably engage the second threaded portion to allow the second threaded portion to threadably pass through the first threaded portion, thereby positioning the first slot between the head portion and the second threaded portion, resulting in the first fastener being captively received in the first slot.

[0042] In the fracture plating system of any preceding paragraph, the first fastener and the first slot may be configured to allow translation of the first fastener along the first longitudinal axis, with the

first fastener captively received in the first slot, and the first fastener may be further configured so that, with the first fastener captively received in the first slot, the first fastener may be moveable relative to the first plate along a third longitudinal axis of the first fastener.
[0043] In the fracture plating system of any preceding paragraph, the fracture plating system may further include a first locking nut having a first body length, and a second locking nut having a second body length, different than the first body length, wherein the first locking nut and the second locking nut may be interchangeably securable to the first fastener and the second fastener.
[0044] In the fracture plating system of any preceding paragraph, the first fastener may include a first fastener length, and the second fastener may include a second fastener length, different than the first fastener length.
[0045] In the fracture plating system of any preceding paragraph, the bone may include a rib, and the fracture plating system may further include a tether configured to draw the first plate and two of the first fastener and the second fastener through a VATS portal to the interior surface of the bone.
[0046] In the fracture plating system of any preceding paragraph, the first fastener may include a first head portion, and the first fastener may be configured to reside captive within the first slot independently of engagement of any protruding feature of the first head portion with the first plate.
[0047] In some embodiments, a fracture plating system may be configured to stabilize a first fracture and a second fracture of a bone of a patient, the bone may include an interior surface facing toward an interior body cavity of the patient, and an exterior surface facing away from the interior body cavity. The fracture plating system may include a plate configured to be placed on the interior surface, the plate having one or more slots and configured to span the first fracture and the second fracture, and four fasteners each configured to be received in the bone and in one of the one or more slots to secure the plate to the interior surface of the bone. The four fasteners may be configured to be received in the bone on opposite sides of each of the first fracture and the second fracture.
[0048] In the fracture plating system of any preceding paragraph, the fracture plating system may further include a first tether configured to guide a first two of the four fasteners to the interior surface of the bone, and a second tether configured to guide a second two of the four fasteners to the interior surface of the bone.
[0049] In the fracture plating system of any preceding paragraph, the bone may include a rib, the first tether may be further configured to draw the plate and a first two of the four fasteners through a VATS portal to the interior surface of the bone, and the second tether may be further configured to draw a second two of the four fasteners through the VATS portal to the interior surface of the bone.
[0050] In the fracture plating system of any preceding paragraph, the first tether may be further configured to draw a first one of the four fasteners through a first hole in a first portion of the bone proximate a first side of one of the first fracture and the second fracture, and draw a second one of the four fasteners through a second hole in a second portion of the bone proximate a second side of one of the first fracture and the second fracture.
[0051] In the fracture plating system of any preceding paragraph, each of the one or more slots may be configured to span at least one of the first fracture and the second fracture.
[0052] In the fracture plating system of any preceding paragraph, each of the one or more slots may be configured to receive at least two of the four fasteners.
[0053] In the fracture plating system of any preceding paragraph, the fracture plating system may further include four locking nuts each configured to receive one of the four fasteners and cooperate with one of the four fasteners to secure the plate to the interior surface of the bone.
[0054] In some embodiments, a fracture plating system may be configured to stabilize a first fracture and a second fracture of a bone of a patient, wherein the first fracture and the second fracture may define a flail segment of the bone, the bone may include an interior surface facing toward an interior body cavity of the patient, and an exterior surface facing away from the interior body cavity. The fracture plating system may include a plate having one or more slots extending

along a longitudinal axis of the plate and configured to span the first fracture and the second fracture, and at least three fasteners configured to be captively received in the one or more slots. A first fastener and a second fastener may be configured to be received in the bone on opposite sides of the first fracture and the second fracture to secure the plate to the interior surface of the bone, and a third fastener may be configured to be received in the flail segment to secure the plate to the flail segment.

[0055] In the fracture plating system of any preceding paragraph, the fracture plating system may further include a tether configured to guide two of the at least three fasteners to the interior surface of the bone.

[0056] In the fracture plating system of any preceding paragraph, the tether may be further configured to guide the plate to the interior surface of the bone.

[0057] In the fracture plating system of any preceding paragraph, the bone may include a rib, and the tether may be further configured to draw the plate and two of the at least three fasteners through a VATS portal to the interior surface of the bone.

[0058] In the fracture plating system of any preceding paragraph, each of the one or more slots may be configured to span at least one of the first fracture and the second fracture.

[0059] In the fracture plating system of any preceding paragraph, the fracture plating system may further include at least three locking nuts configured to receive one of the at least three fasteners and cooperate with one of the at least three fasteners to secure the plate to the interior surface of the bone.

[0060] In some embodiments, a fracture plating system may be configured to stabilize multiple fractures of a bone of a patient, wherein the multiple fractures may define one or more flail segments, the bone may include an interior surface facing toward an interior body cavity of the patient, and an exterior surface facing away from the interior body cavity. The fracture plating system may include a plate having one or more slots extending along a longitudinal axis of the plate and configured to span the first fracture and the second fracture, and at least three fasteners configured to be captively received in the one or more slots. A first fastener and a second fastener may be configured to be received in the bone on opposite sides of the first fracture and the second fracture to secure the plate to the interior surface of the bone, and a third fastener may be configured to be received in the flail segment to secure the plate to the flail segment.

[0061] In the fracture plating system of any preceding paragraph, one of the one or more tethers may be further configured to guide the plate to the interior surface of the bone.

[0062] In the fracture plating system of any preceding paragraph, each of the one or more slots may be configured to span at least one of the multiple fractures.

[0063] In the fracture plating system of any preceding paragraph, the plurality of fasteners may be configured to be received in the bone on opposite sides of the multiple fractures to secure the plate to the interior surface of the bone.

[0064] In the fracture plating system of any preceding paragraph, the bone may include a rib, and one of the one or more tethers may be further configured to draw the plate and two of the plurality of fasteners through a VATS portal to the interior surface of the bone.

[0065] In the fracture plating system of any preceding paragraph, the fracture plating system may further include a plurality of locking nuts configured to receive one of the plurality of fasteners and cooperate with one of the plurality of fasteners to secure the plate to the interior surface of the bone.

[0066] In the fracture plating system of any preceding paragraph, each of the one or more slots may be configured to receive at least two of the plurality of fasteners.

[0067] In some embodiments, a fracture plating system may be configured to stabilize a first fracture of a bone of a patient, the bone may include an interior surface facing toward an interior body cavity of the patient, and an exterior surface facing away from the interior body cavity. The fracture plating system may include a plate having one or more slots and configured to span the

first fracture, two or more fasteners configured to be received in the one or more slots, and one or more tethers configured to guide the two or more fasteners to the interior surface of the bone. Each of the two or more fasteners may be configured to be received in a hole in the bone, and the two or more fasteners may be configured to secure the plate to the exterior surface of the bone.
[0068] In the fracture plating system of any preceding paragraph, the fracture plating system may further include two or more locking nuts configured to receive the two or more fasteners, wherein the two or more locking nuts may be configured to cooperate with the two or more fasteners to secure the plate to the exterior surface of the bone.
[0069] In the fracture plating system of any preceding paragraph, the two or more fasteners may each include a proximal end comprising a head portion, and a distal end opposite the proximal end, and, with the plate secured to the exterior surface of the bone, the two or more locking nuts may engage the plate and the head portion may contact the interior surface of the bone.
[0070] In the fracture plating system of any preceding paragraph, the bone may include a rib, and the one or more tethers may be further configured to draw the two or more fasteners through a VATS portal to the interior surface of the bone.
[0071] In the fracture plating system of any preceding paragraph, the two or more fasteners may each include a proximal end comprising a head portion, and a distal end opposite the proximal end, and, with the plate secured to the exterior surface of the bone, the head portion may not contact the plate.
[0072] In the fracture plating system of any preceding paragraph, the head portion may contact the interior surface and the distal end may extend from the exterior surface of the bone.
[0073] In the fracture plating system of any preceding paragraph, the one or more tethers may be further configured to guide the two or more fasteners to the one or more slots.
[0074] In some embodiments, a fracture plating system may be configured to stabilize a first fracture of a bone of a patient, the bone may include an interior surface facing toward an interior body cavity of the patient, and an exterior surface facing away from the interior body cavity. The fracture plating system may include a plate having one or more slots and configured to span the first fracture, and two or more fasteners configured to secure the plate to the exterior surface of the bone. Each of the two or more fasteners may include a proximal end having a head portion, and a distal end opposite the proximal end, the distal end may be configured to be received in a hole in the bone, the distal end may be further configured to be received in one of the one or more slots, and, with the plate secured to the exterior surface of the bone, the head portion may not contact the plate.
[0075] In the fracture plating system of any preceding paragraph, the fracture plating system may further include two or more locking nuts configured to receive the two or more fasteners, wherein the two or more locking nuts may be configured to cooperate with the two or more fasteners to secure the plate to the exterior surface of the bone.
[0076] In the fracture plating system of any preceding paragraph, with the plate secured to the exterior surface of the bone, the two or more locking nuts may engage the plate.
[0077] In the fracture plating system of any preceding paragraph, with the plate secured to the exterior surface of the bone, the head portion may contact the interior surface of the bone.
[0078] In the fracture plating system of any preceding paragraph, the fracture plating system may further include a tether configured to guide the two or more fasteners to the one or more slots.
[0079] In the fracture plating system of any preceding paragraph, the bone may include a rib, and the tether may be further configured to draw the two or more fasteners through a VATS portal to the interior surface of the bone.
[0080] In the fracture plating system of any preceding paragraph, the head portion may contact the interior surface and the distal end may extend from the exterior surface of the bone.
[0081] In some embodiments, a fracture plating system may be configured to stabilize a first fracture of a bone of a patient, the bone may include an interior surface facing toward an interior

body cavity of the patient, and an exterior surface facing away from the interior body cavity. The fracture plating system may include a plate having one or more slots and configured to span the first fracture, two or more fasteners each having a proximal end having a head portion, and a distal end opposite the proximal end, and two or more locking nuts configured to be received on the distal end. Each of the two or more fasteners may be configured to be received in a hole in the bone so that the head portion may contact the interior surface and the distal end may extend from the exterior surface of the bone, and the two or more locking nuts may be configured to cooperate with the two or more fasteners to secure the plate to the exterior surface of the bone.
[0082] In the fracture plating system of any preceding paragraph, the fracture plating system may further include a tether configured to guide the two or more fasteners to the interior surface of the bone.
[0083] In the fracture plating system of any preceding paragraph, the bone may include a rib, and the tether may be further configured to draw the two or more fasteners through a VATS portal to the interior surface of the bone.
[0084] In the fracture plating system of any preceding paragraph, the tether may be further configured to guide the two or more fasteners to the one or more slots.
[0085] In the fracture plating system of any preceding paragraph, each of the two or more fasteners may be configured to be received in a hole in the bone.
[0086] In the fracture plating system of any preceding paragraph, with the plate secured to the exterior surface of the bone, the head portion may not contact the plate.
[0087] These and other features and advantages of the present disclosure will become more fully apparent from the following description and appended claims or may be learned by the practice of the implants, systems, and methods set forth hereinafter.

Description

BRIEF DESCRIPTION OF THE DRA WINGS
[0088] Exemplary embodiments of the present disclosure will become more fully apparent from the following description taken in conjunction with the accompanying drawings. Understanding that these drawings depict only exemplary embodiments and are, therefore, not to be considered limiting of the scope of the present disclosure, the exemplary embodiments of the present disclosure will be described with additional specificity and detail through use of the accompanying drawings.
[0089] FIG. **1** is a perspective view of a fracture plating system according to an embodiment of the present disclosure.
[0090] FIG. **2**A is a front view of a plate of the fracture plating system of FIG. **1** according to an embodiment of the present disclosure.
[0091] FIG. **2**B is a bottom view of the plate of FIG. **2**A.
[0092] FIG. **3**A is a front view of an anti-rotation fastener of the fracture plating system of FIG. **1** according to an embodiment of the present disclosure.
[0093] FIG. **3**B is a side view of the anti-rotation fastener of FIG. **3**A.
[0094] FIG. **4**A is a front view of a circular head fastener of the fracture plating system of FIG. **1** according to an embodiment of the present disclosure.
[0095] FIG. **4**B is a side view of the circular head fastener of FIG. **4**A.
[0096] FIG. **5**A is a perspective view of a locking nut of the fracture plating system of FIG. **1** according to an embodiment of the present disclosure.
[0097] FIG. **5**B is a top view of the locking nut of FIG. **5**A.
[0098] FIG. **5**C is a side view of the locking nut of FIG. **5**A.
[0099] FIG. **6**A is a perspective view of a washer of the fracture plating system of FIG. **1** according

to an embodiment of the present disclosure.
[0100] FIG. **6**B is a top view of the washer of FIG. **6**A.
[0101] FIG. **6**C is a side view of the washer of FIG. **6**A.
[0102] FIG. **7** is a perspective view of the fracture plating system of FIG. **1** in a lower profile configuration.
[0103] FIG. **8** is a perspective view of the fracture plating system of FIG. **1** spanning an exemplary bone fracture.
[0104] FIG. **9** is a perspective view of the fracture plating system of FIG. **1** in a lower profile configuration.
[0105] FIG. **10**A is a perspective view of the plate of the fracture plating system of FIG. **1** according to an embodiment of the present disclosure.
[0106] FIG. **10**B is a top perspective partial view of the plate of FIG. **10**A.
[0107] FIG. **11** is a front view of the fracture plating system of FIG. **1** in a first pre-assembled configuration.
[0108] FIG. **12** is a front view of the fracture plating system of FIG. **1** in a second pre-assembled configuration.
[0109] FIG. **13** is a front partial section view of the fracture plating system of FIG. **12**.
[0110] FIG. **14** is a front section view of the fracture plating system of FIG. **12** in a third pre-assembled configuration.
[0111] FIG. **15** is a front view of the fracture plating system of FIG. **1** in an assembled configuration.
[0112] FIG. **16** is a front section view of the fracture plating system of FIG. **15**.
[0113] FIG. **17** is a front section view of the fracture plating system of FIG. **1** in a lower profile configuration.
[0114] FIG. **18** is a perspective view of the fracture plating system of FIG. **1** in a partially deployed configuration spanning an exemplary bone fracture.
[0115] FIG. **19** is a front section view of the fracture plating system of FIG. **18** in a partially deployed configuration spanning an exemplary bone fracture.
[0116] FIG. **20** is a perspective view of the fracture plating system of FIG. **18** in a partially deployed configuration spanning an exemplary bone fracture.
[0117] FIG. **21** is a front section view of the fracture plating system of FIG. **20** in a partially deployed configuration spanning an exemplary bone fracture.
[0118] FIG. **22** is a front section view of the fracture plating system of FIG. **20** in a partially deployed configuration spanning an exemplary bone fracture.
[0119] FIG. **23** is a perspective view of the fracture plating system of FIG. **20** in a partially deployed configuration spanning an exemplary bone fracture.
[0120] FIG. **24** is a perspective view of the fracture plating system of FIG. **1** in a partially deployed configuration spanning an exemplary bone fracture.
[0121] FIG. **25** is a front view of the fracture plating system of FIG. **24** in a partially deployed configuration spanning an exemplary bone fracture.
[0122] FIG. **26** is a perspective view of the fracture plating system of FIG. **1** in a partially deployed configuration spanning an exemplary bone fracture.
[0123] FIG. **27** is a perspective view of the fracture plating system of FIG. **26** in a partially deployed configuration spanning an exemplary bone fracture.
[0124] FIG. **28** is a perspective view of the fracture plating system of FIG. **26** in a deployed configuration spanning an exemplary bone fracture.
[0125] FIG. **29** is a perspective view of a fracture plating system according to an embodiment of the present disclosure.
[0126] FIG. **30** is a perspective section view of the fracture plating system of FIG. **29**.
[0127] FIG. **31** is a partial bottom perspective view of the fracture plating system of FIG. **29**.

[0128] FIG. **32**A is a top view of a circular head fixed hinge fastener of the fracture plating system of FIG. **29** according to an embodiment of the present disclosure.
[0129] FIG. **32**B is a front view of the circular head fixed hinge fastener of FIG. **32**A.
[0130] FIG. **32**C is a side view of the circular head fixed hinge fastener of FIG. **32**A.
[0131] FIG. **33**A is a top view of a fixed hinge fastener of the fracture plating system of FIG. **29** according to an embodiment of the present disclosure.
[0132] FIG. **33**B is a front view of the fixed hinge fastener of FIG. **33**A.
[0133] FIG. **33**C is a side view of the fixed hinge fastener of FIG. **33**A.
[0134] FIG. **34** is a perspective view of a fracture plating system according to an embodiment of the present disclosure.
[0135] FIG. **35** is a bottom perspective view of the fracture plating system of FIG. **34** in a pre-assembled configuration.
[0136] FIG. **36**A is a front view of a plate of the fracture plating system of FIG. **34** according to an embodiment of the present disclosure.
[0137] FIG. **36**B is a bottom view of the plate of FIG. **36**A.
[0138] FIG. **37**A is a front view of a fixed hinge fastener of the fracture plating system of FIG. **34** according to an embodiment of the present disclosure.
[0139] FIG. **37**B is a side view of the fixed hinge fastener of FIG. **37**A.
[0140] FIG. **38**A is a front view of the anti-rotation fastener of FIG. **3**A.
[0141] FIG. **38**B is a side view of the anti-rotation fastener of FIG. **38**A.
[0142] FIG. **39** is a perspective view of a fracture plating system according to an embodiment of the present disclosure.
[0143] FIG. **40** is a bottom perspective view of the fracture plating system of FIG. **39**.
[0144] FIG. **41**A is a front view of a plate of the fracture plating system of FIG. **39** according to an embodiment of the present disclosure.
[0145] FIG. **41**B is a bottom view of the plate of FIG. **41**A.
[0146] FIG. **42**A is a front view of the circular head fastener of FIG. **4**A.
[0147] FIG. **42**B is a side view of the circular head fastener of FIG. **42**A
[0148] FIG. **43**A is a front view of a ball-headed fastener of the fracture plating system of FIG. **39** according to an embodiment of the present disclosure.
[0149] FIG. **43**B is a side view of the ball-headed fastener of FIG. **43**A.
[0150] FIG. **44**A is a perspective view of a fracture plating system according to an embodiment of the present disclosure.
[0151] FIG. **44**B is a partial perspective view of the fracture plating system of FIG. **44**A.
[0152] FIG. **45** is a bottom perspective view of the fracture plating system of FIG. **44**A.
[0153] FIG. **46**A is a top view of a plate of the fracture plating system of FIG. **44**A according to an embodiment of the present disclosure.
[0154] FIG. **46**B is a front view of the plate of FIG. **46**A.
[0155] FIG. **47**A is a top view of a pin fastener of the fracture plating system of FIG. **44**A according to an embodiment of the present disclosure.
[0156] FIG. **47**B is a front view of the pin fastener of FIG. **47**A.
[0157] FIG. **47**C is a side view of the pin fastener of FIG. **47**A.
[0158] FIG. **48**A is a front view of a fracture plating system in a partially deployed configuration according to an embodiment of the present disclosure spanning a plurality of exemplary bone fractures.
[0159] FIG. **48**B is a perspective view of a fracture plating system secured to a bone with multiple fractures according to an embodiment of the present disclosure.
[0160] FIG. **48**C is a perspective view of the fracture plating system of FIG. **48**B.
[0161] FIG. **48**D is a perspective view of a fracture plating system secured to a bone with multiple fractures according to an embodiment of the present disclosure.

[0162] FIG. **48**E is a perspective view of the fracture plating system of FIG. **48**D.
[0163] FIG. **49** is a front view of a fracture plating system in a partially deployed configuration according to an embodiment of the present disclosure spanning a plurality of exemplary bone fractures.
[0164] FIG. **50** is a bottom perspective view of the fracture plating system of FIG. **49** in a partially deployed configuration spanning a plurality of exemplary bone fractures.
[0165] FIG. **51** is a bottom perspective view of a fracture plating system in a partially deployed configuration according to an embodiment of the present disclosure spanning an exemplary bone fracture.
[0166] FIG. **52** is a partial bottom perspective view of the fracture plating system of FIG. **51** spanning an exemplary bone fracture.
[0167] FIG. **53** is a front view of a fracture plating system in a partially deployed configuration according to an embodiment of the present disclosure spanning an exemplary bone fracture.
[0168] FIG. **54** is a partial front view of the fracture plating system of FIG. **53** spanning an exemplary bone fracture.
[0169] FIG. **55** is a partial bottom perspective view of the fracture plating system of FIG. **53** spanning an exemplary bone fracture.
[0170] FIG. **56** is a front view of a fracture plating system according to an embodiment of the present disclosure spanning a plurality of exemplary bone fractures.
[0171] FIG. **57** is a bottom perspective view of the fracture plating system of FIG. **56** spanning a plurality of exemplary bone fractures.
[0172] FIG. **58**A is a front view of a plate of the fracture plating system of FIG. **56** according to an embodiment of the present disclosure.
[0173] FIG. **58**B is a bottom view of the plate of FIG. **58**A.
[0174] FIG. **59** is a perspective view of a fracture plating system according to an embodiment of the present disclosure spanning a plurality of exemplary bone fractures.
[0175] FIG. **60** is a perspective view of the fracture plating system of FIG. **59** spanning a plurality of exemplary bone fractures.
[0176] FIG. **61** is a bottom perspective view of the fracture plating system of FIG. **59** spanning a plurality of exemplary bone fractures.
[0177] FIG. **62**A is a front view of a plate of the fracture plating system of FIG. **59** according to an embodiment of the present disclosure.
[0178] FIG. **62**B is bottom view of the plate of FIG. **62**A.
[0179] FIG. **63**A is a front view of a threaded fastener of the fracture plating system of FIG. **59** according to an embodiment of the present disclosure.
[0180] FIG. **63**B is a side view of the threaded fastener of FIG. **63**A.
[0181] FIG. **64**A front view of a toggle fastener in an insertion configuration of a fracture plating system according to an embodiment of the present disclosure.
[0182] FIG. **64**B is a front view of the toggle fastener of FIG. **64**A in a deployed configuration.
[0183] FIG. **65** is a perspective view of a pair of toggle fasteners of FIG. **64**A in a partially deployed configuration spanning an exemplary bone fracture.
[0184] FIG. **66** is a perspective view of a pair of toggle fasteners of FIG. **64**A in a partially deployed configuration spanning an exemplary bone fracture.
[0185] FIG. **67** is a perspective view of a pair of toggle fasteners of FIG. **64**A in a partially deployed configuration spanning an exemplary bone fracture.
[0186] FIG. **68** is a perspective view of a pair of toggle fasteners of FIG. **64**A in a partially deployed configuration spanning an exemplary bone fracture.
[0187] FIG. **69** is a perspective view of a fracture plating system including the toggle fastener of FIG. **64**A according to an embodiment of the present disclosure spanning an exemplary bone fracture.

[0188] FIG. **70** is a bottom perspective view of the fracture plating system of FIG. **69** spanning an exemplary bone fracture.
[0189] FIG. **71** is a bottom view of the plate of FIG. **2**A.
[0190] FIG. **72** is a perspective view of a fracture plating system according to an embodiment of the present disclosure in a partially assembled configuration.
[0191] FIG. **73** is a perspective view of the fracture plating system of FIG. **72** in a partially assembled configuration.
[0192] FIG. **74**A is a front view of a post fastener of the fracture plating system of FIG. **72** according to an embodiment of the present disclosure.
[0193] FIG. **74**B is a side view of the post fastener of FIG. **74**A.
[0194] FIG. **75**A is a front view of a nut of the fracture plating system of FIG. **72** according to an embodiment of the present disclosure.
[0195] FIG. **75**B is a side view of the nut of FIG. **75**A.
[0196] FIG. **75**C is a perspective view of the nut of FIG. **75**A.
[0197] FIG. **76**A is top view of a pin fastener of a fracture plating system of FIG. **81** according to an embodiment of the present disclosure.
[0198] FIG. **76**B is a perspective view of the pin fastener of FIG. **76**A.
[0199] FIG. **76**C is a front view of the pin fastener of FIG. **76**A.
[0200] FIG. **76**D is a side view of the pin fastener of FIG. **76**A.
[0201] FIG. **77**A is a top view of a plate of the fracture plating system of FIG. **81** according to an embodiment of the present disclosure.
[0202] FIG. **77**B is a front view of the plate of FIG. **77**A.
[0203] FIG. **78** is a partial top view of the fracture plating system of FIG. **81** in a partially assembled configuration.
[0204] FIG. **79** is a partial perspective view of the fracture plating system of FIG. **81** in a partially assembled configuration.
[0205] FIG. **80** is a perspective section view of the fracture plating system of FIG. **81**.
[0206] FIG. **81** is a perspective view of a fracture plating system according to an embodiment of the present disclosure.
[0207] FIG. **82** is a partial perspective view of the fracture plating system of FIG. **81**.
[0208] FIG. **83** is a partial perspective view of the fracture plating system of FIG. **81**.
[0209] FIG. **84** is a perspective view of an exemplary rig cage with an exemplary bone fracture.
[0210] FIG. **85** is a partial perspective view of the exemplary rib cage of FIG. **84**.
[0211] FIG. **86** is a partial perspective view of the exemplary rib cage of FIG. **84** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.
[0212] FIG. **87** is a partial perspective view of the exemplary rib cage of FIG. **84** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.
[0213] FIG. **88** is a partial perspective view of the exemplary rib cage of FIG. **84** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.
[0214] FIG. **89**A is a front view of a set of differently sized plates according to an embodiment of the present disclosure.
[0215] FIG. **89**B is a front view of a set of differently sized fasteners according to an embodiment of the present disclosure.
[0216] FIG. **89**C is a front view of a set of differently sized locking nuts according to an embodiment of the present disclosure.
[0217] FIG. **89**D Is a partial perspective view of an exemplary rib cage showing an exemplary fracture.

[0218] FIG. **90** is a front view of a fracture plating system illustrating a step in a method of deploying the fracture plating system according to an embodiment of the present disclosure.
[0219] FIG. **91** is a front view of the fracture plating system of FIG. **90** illustrating a step in a method of deploying the fracture plating system according to an embodiment of the present disclosure.
[0220] FIG. **92** is a front view of the fracture plating system of FIG. **90** illustrating a step in a method of deploying the fracture plating system according to an embodiment of the present disclosure.
[0221] FIG. **93** is a partial perspective view of the exemplary rib cage of FIG. **84** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.
[0222] FIG. **94** is a partial perspective view of the exemplary rib cage of FIG. **84** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.
[0223] FIG. **95** is a partial perspective view of the exemplary rib cage of FIG. **84** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.
[0224] FIG. **96** is a partial perspective view of the exemplary rib cage of FIG. **84** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.
[0225] FIG. **97** is a partial perspective view of the exemplary rib cage of FIG. **84** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.
[0226] FIG. **98** is a partial perspective view of the exemplary rib cage of FIG. **84** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.
[0227] FIG. **99** is a partial perspective view of the exemplary rib cage of FIG. **84** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.
[0228] FIG. **100** is a partial perspective view of the exemplary rib cage of FIG. **84** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.
[0229] FIG. **101** is a partial perspective view of the exemplary rib cage of FIG. **84** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.
[0230] FIG. **102** is a perspective view of a fracture plating system according to an embodiment of the present disclosure spanning an exemplary bone fracture.
[0231] FIG. **103** is a perspective view of a fracture plating system of FIG. **102**.
[0232] FIG. **104**A is a front view of an anti-rotation fastener of the fracture plating system of FIG. **102** according to an embodiment of the present disclosure.
[0233] FIG. **104**B is a side view of the anti-rotation fastener of FIG. **104**A.
[0234] FIG. **105**A is a front view of a locking nut of the fracture plating system of FIG. **102** according to an embodiment of the present disclosure.
[0235] FIG. **105**B is a side view of the locking nut of FIG. **105**A.
[0236] FIG. **106**A is a perspective view of a driver of the fracture plating system of FIG. **102** according to an embodiment of the present disclosure.
[0237] FIG. **106**B is a side view of the driver of FIG. **106**A.
[0238] FIG. **106**C is a front view of the driver of FIG. **106**A.
[0239] FIG. **107** is a front perspective view of a spinal fixation plating system, secured to an exemplary portion of a spine, according to an embodiment of the present disclosure.

[0240] FIG. **108** is a rear perspective view of the spinal fixation plating system of FIG. **107** secured to an exemplary portion of a spine.
[0241] FIG. **109** is a front perspective view of the spinal fixation plating system of FIG. **107** secured to an exemplary portion of a spine.
[0242] FIG. **110** is a rear perspective view of the spinal fixation plating system of FIG. **107** secured to an exemplary portion of a spine.
[0243] FIG. **111**A is a front view of a plate of the spinal fixation plating system of FIG. **107** according to an embodiment of the present disclosure.
[0244] FIG. **111**B is a bottom view of the plate of FIG. **111**A.
[0245] FIG. **112**A is a perspective view of a locking nut of the spinal fixation plating system of FIG. **107** according to an embodiment of the present disclosure.
[0246] FIG. **112**B is a front view of the locking nut of FIG. **112**A.
[0247] FIG. **112**C is a side view of the locking nut of FIG. **112**A.
[0248] FIG. **113**A is a perspective view of an anti-rotation fastener of the spinal fixation plating system of FIG. **107** according to an embodiment of the present disclosure.
[0249] FIG. **113**B is a front view of the anti-rotation fastener of FIG. **113**A.
[0250] FIG. **113**C is a side view of the anti-rotation fastener of FIG. **113**A.
[0251] FIG. **114** is a front view of a fracture plating system according to an embodiment of the present disclosure spanning an exemplary bone fracture.
[0252] FIG. **115** is a partial perspective view of the fracture plating system of FIG. **114**.
[0253] FIG. **116** is a partial section view of the fracture plating system of FIG. **114**.
[0254] FIG. **117** is a front section view of the fracture plating system of FIG. **114**.
[0255] FIG. **118**A is a bottom perspective view of a ratchet cap of the fracture plating system of FIG. **114** according to an embodiment of the present disclosure.
[0256] FIG. **118**B is front perspective view of the ratchet cap of FIG. **118**A.
[0257] FIG. **118**C is a front view of the ratchet cap of FIG. **118**A.
[0258] FIG. **118**D is a side view of the ratchet cap of FIG. **118**A.
[0259] FIG. **119**A is a perspective view of a ratchet fastener of the fracture plating system of FIG. **114** according to an embodiment of the present disclosure.
[0260] FIG. **119**B is a front view of the ratchet fastener of FIG. **119**A.
[0261] FIG. **119**C is a side view of the ratchet fastener of FIG. **119**A.
[0262] FIG. **120**A is a perspective view of a fracture repair system in an undeformed configuration according to an embodiment of the present disclosure.
[0263] FIG. **120**B is a front view of the fracture repair system of FIG. **120**A in an undeformed configuration.
[0264] FIG. **120**C is a side view of the fracture repair system of FIG. **120**A in an undeformed configuration.
[0265] FIG. **121** is a front view of the fracture repair system of FIG. **120**A in an undeformed configuration.
[0266] FIG. **122** is a front view of the fracture repair system of FIG. **120**A in an expanded configuration according to an embodiment of the present disclosure.
[0267] FIG. **123** is a front section view of an exemplary rib cage with the fracture repair system of FIG. **122** spanning exemplary fractures.
[0268] FIG. **124** is a front section view of an exemplary rib cage with the fracture repair system of FIG. **120**A compressing exemplary fractures.
[0269] FIG. **125** is a front section view of an exemplary rib cage with the fracture repair system of FIG. **120**A compressing exemplary fractures.
[0270] FIG. **126**A is a perspective view of an application instrument of a fracture repair system according to an embodiment of the present disclosure.
[0271] FIG. **126**B is a side view of the application instrument of FIG. **126**A.

[0272] FIG. **126**C is a front view of the application instrument of FIG. **126**A.
[0273] FIG. **127** is a balloon of a fracture repair system according to an embodiment of the present disclosure.
[0274] FIG. **128** is a partial perspective view of an exemplary rib cage with an exemplary bone fracture.
[0275] FIG. **129** is a partial perspective view of the exemplary rib cage of FIG. **128** illustrating a step in a method of deploying a fracture repair system according to an embodiment of the present disclosure.
[0276] FIG. **130** is a partial perspective view of the exemplary rib cage of FIG. **128** illustrating a step in a method of deploying a fracture repair system according to an embodiment of the present disclosure.
[0277] FIG. **131** is a partial perspective view of the exemplary rib cage of FIG. **128** illustrating a step in a method of deploying a fracture repair system according to an embodiment of the present disclosure.
[0278] FIG. **132** is a partial perspective view of the exemplary rib cage of FIG. **128** illustrating a step in a method of deploying a fracture repair system according to an embodiment of the present disclosure.
[0279] FIG. **133** is a partial perspective view of the exemplary rib cage of FIG. **128** illustrating a step in a method of deploying a fracture repair system according to an embodiment of the present disclosure.
[0280] FIG. **134** is a partial perspective view of the exemplary rib cage of FIG. **128** illustrating a step in a method of deploying a fracture repair system according to an embodiment of the present disclosure.
[0281] FIG. **135** is a partial perspective section view of the exemplary rib cage of FIG. **128** illustrating a step in a method of deploying a fracture repair system according to an embodiment of the present disclosure.
[0282] FIG. **136** is a partial perspective view of the exemplary rib cage of FIG. **128** illustrating a step in a method of deploying a fracture repair system according to an embodiment of the present disclosure.
[0283] It is to be understood that the drawings are for purposes of illustrating the concepts of the present disclosure and may not be drawn to scale. Furthermore, the drawings illustrate exemplary embodiments and do not represent limitations to the scope of the present disclosure.

DETAILED DESCRIPTION

[0284] Exemplary embodiments of the present disclosure will be best understood by reference to the drawings, wherein like parts are designated by like numerals throughout. It will be readily understood that the components of the present disclosure, as generally described and illustrated in the drawings, could be arranged, and designed in a wide variety of different configurations. Thus, the following more detailed description of the embodiments of the devices, systems, and methods, as represented in the drawings, is not intended to limit the scope of the present disclosure but is merely representative of exemplary embodiments of the present disclosure.
[0285] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments. While the various aspects of the embodiments are presented in the drawings, the drawings are not necessarily drawn to scale unless specifically indicated.
[0286] Standard medical planes of reference and descriptive terminology are employed in this specification. While these terms are commonly used to refer to the human body, certain terms are applicable to physical objects in general.
[0287] A standard system of three mutually perpendicular reference planes is employed. A sagittal plane divides a body into right and left portions. A coronal plane divides a body into anterior and

posterior portions. A transverse plane divides a body into superior and inferior portions. A mid-sagittal, mid-coronal, or mid-transverse plane divides a body into equal portions, which may be bilaterally symmetric. The intersection of the sagittal and coronal planes defines a superior-inferior or cephalad-caudal axis. The intersection of the sagittal and transverse planes defines an anterior-posterior axis. The intersection of the coronal and transverse planes defines a medial-lateral axis. The superior-inferior or cephalad-caudal axis, the anterior-posterior axis, and the medial-lateral axis are mutually perpendicular.

[0288] Anterior means toward the front of a body. Posterior means toward the back of a body. Superior or cephalad means toward the head. Inferior or caudal means toward the feet or tail. Medial means toward the midline of a body, particularly toward a plane of bilateral symmetry of the body. Lateral means away from the midline of a body or away from a plane of bilateral symmetry of the body. Axial means toward a central axis of a body. Abaxial means away from a central axis of a body. Ipsilateral means on the same side of the body. Contralateral means on the opposite side of the body. Proximal means toward the trunk of the body. Proximal may also mean toward a user or operator. Distal means away from the trunk. Distal may also mean away from a user or operator. Dorsal means toward the top of the foot. Plantar means toward the sole of the foot. Varus means deviation of the distal part of the leg below the knee inward, resulting in a bowlegged appearance. Valgus means deviation of the distal part of the leg below the knee outward, resulting in a knock-kneed appearance.

[0289] The present disclosure relates to fracture plating devices, systems, and methods. Those skilled in the art will recognize that the following description is merely illustrative of the principles of the technology, which may be applied in various ways to provide many alternative embodiments. The present disclosure illustrates devices for plating systems for one or more fractures of a rib for the purposes of illustrating the concepts of the present design. However, it will be understood that other variations and uses are contemplated including, but not limited to, fractures of metatarsals, fractures of phalanges, metacarpals, and carpals, fractures of a fibula, fractures of an ulna, and other bone fractures.

[0290] FIG. **1** is a perspective view of a fracture plating system **100** according to an embodiment of the present disclosure. The fracture plating system **100** may be configured to stabilize bone fractures through intra-thoracic plating. The fracture plating system **100** may be configured to be secured to an interior surface of a bone, an exterior surface of a bone, or both an interior surface of a bone and an exterior surface of a bone. The fracture plating system **100** may beneficially require a smaller incision in the chest of a patient as compared to a plating system that is only configured to be secured to an exterior surface of a bone. The fracture plating system **100** may be configured to be introduced into an intra-thoracic cavity, or interior cavity, through one or more portals, commonly used for Video-assisted Thoracic Surgery (VATS) and/or a thoracoscopic procedure. Additionally, or alternatively, the fracture plating system **100** may be configured to be introduced into an intra-thoracic cavity, or interior body cavity, using a trans-intercostal approach. The trans-intercostal approach may include a surgical technique that involves making incisions between two adjacent ribs to access the thoracic cavity, or interior cavity.

[0291] The fracture plating system **100** may include a plate and a plurality of fasteners. The plates may be one of a set of differently sized implants. The fasteners may be of a set of differently sized implants. The fracture plating system **100** may be modular in that a specific sized plate may be chosen by a user based on patient anatomy and quantity and location of fractures. Additionally, a plurality of fasteners may be selected by a user based on patient anatomy and quantity and location of fractures. The selection of fasteners may include fasteners of the same or different lengths and/or diameters. After a selection is made, the plate and plurality of fasteners may be assembled by a user, for example, during a surgical procedure, to best address a specific patient's indications. The fracture plating system **100** may be configured to achieve the fixation of a fracture plate to a portion of bone using a single tether to pull the plate against the bone.

[0292] The fracture plating system **100** may include a plate **102**, an anti-rotation fastener **150** and/or a circular head fastener **180**, a locking nut **30**, and a washer **50**. FIG. **2**A is a front view of a plate **102** of the fracture plating system **100** according to an embodiment of the present disclosure. FIG. **2**B is a bottom view of the plate **102**. The plate **102** may be configured to stabilize and/or facilitate reduction of a fracture as a component of the fracture plating system **100**. The plate **102** may include a plate length **103** and a plate radius **104**. The plate length **103** may be configured to span one or more fractures.

[0293] The plate **102** may be configured such that the plate length **103** is within a range of lengths from 30 mm to 300 mm. The plate **102** may be one of a set of differently-sized implants, each having a different plate length **103**. The plate radius **104** may generally match a contour of an interior surface of a rib. The plate **102** may further be one of a set of differently-sized implants, each having a different plate radius **104**.

[0294] The plate **102** may further include a central portion **125** and a first slot **105** extending along a longitudinal axis of the plate **102** and having a first slot width **107**, a first slot length **108**, and a first threaded feature **110**. The plate **102** may also include a second slot **115** extending along a longitudinal axis of the plate **102** and having a second slot width **117**, a second slot length **118**, and a second threaded feature **120**. The central portion **125** may be generally in the center of the plate **102** and may separate the first slot **105** and the second slot **115**.

[0295] The plate **102** may be fabricated from titanium alloy, titanium, stainless steel, cobalt-chrome, PEEK, PEAK, UHMWPE, a resorbable polymer, or any other biocompatible material with sufficient tensile strength.

[0296] The first slot width **107** and the second slot width **117** may each be configured to receive an anti-rotation fastener **150** and/or a circular head fastener **180**. The first slot width **107** and the second slot width **117** may further be configured to allow translation of the anti-rotation fastener **150** and/or circular head fastener **180** along the length of the first slot and the second slot respectively.

[0297] FIG. **3**A is a front view of an anti-rotation fastener **150** of the fracture plating system **100** according to an embodiment of the present disclosure. FIG. **3**B is a side view of the anti-rotation fastener **150**. The anti-rotation fastener **150** may be configured to secure the plate **102** to a bone portion. The anti-rotation fastener **150** may also be configured to be captively received within the plate **102** while still being allow to translate along the length of the first slot **105** and/or the second slot **115**. The anti-rotation fastener **150** may be configured so that, when a head portion **155** engages the first slot **105** and/or the second slot **115**, the anti-rotation fastener **150** is prevented from rotating in relation to the plate **102**.

[0298] The anti-rotation fastener **150** may include a shank portion **157**, a threaded portion **160**, a tip portion **162**, a cannulation **165**, and a rounded edge **168**. The anti-rotation fastener **150** may further include a proximal end **153** including a head portion **155**, and a distal end **154** opposite the proximal end **153**. The head portion **155** may be configured to be received within the first slot **105** and/or the second slot **115**. The head portion **155** may lack protruding features that are configured to engage with the plate **102**. The threaded portion **160** may be larger than a width of the first slot **105** and/or the second slot **115** to prevent the anti-rotation fastener **150** from disengaging from the plate **102**. The first threaded feature **110** and the second threaded feature **120** may each be configured to threadably engage the threaded portion **160** to allow the threaded portion **160** to pass through the first threaded feature **110** and the second threaded feature **120** respectively.

[0299] The shank portion **157** may be smaller than a width of the first slot **105** and/or the second slot **115** to allow the anti-rotation fastener **150** to translate along the length of the first slot **105** and/or the second slot **115** respectively. The tip portion **162** may be configured to ease insertion of the anti-rotation fastener **150** into a hole in a bone portion. The cannulation **165** may be configured to slidably receive a tether **60**. The rounded edge **168** may reduce stress on the tether **60** when the tether **60** exerts a force on the anti-rotation fastener **150**.

[0300] FIG. **4**A is a front view of a circular head fastener **180** of the fracture plating system **100** according to an embodiment of the present disclosure. FIG. **4**B is a side view of the circular head fastener **180**. The circular head fastener **180** may be configured to secure the plate **102** to a bone portion. The circular head fastener **180** may also be configured to be captively received within the plate **102** while still being allow to translate along the length of the first slot **105** and/or the second slot **115**. The circular head fastener **180** may be configured so that, when a head portion **185** engages the first slot **105** and/or the second slot **115**, the circular head fastener **180** is not prevented from rotating in relation to the plate **102**.

[0301] The circular head fastener **180** may include a shank portion **187**, a threaded portion **190**, a tip portion **192**, a cannulation **195**, and a rounded edge **198**. The circular head fastener **180** may include a proximal end **183** including a head portion **185**, and a distal end **184** opposite the proximal end **183**. The head portion **185** may be configured to be received within the first slot **105** and/or the second slot **115**. The threaded portion **190** may be larger than a width of the first slot **105** and/or the second slot **115** to prevent the circular head fastener **180** from disengaging from the plate **102**.

[0302] The shank portion **187** may be smaller than a width of the first slot **105** and/or the second slot **115** to allow the circular head fastener **180** to translate along the length of the first slot **105** and/or the second slot **115** respectively. The tip portion **192** may be configured to ease insertion of the circular head fastener **180** into a hole in a bone portion. The cannulation **195** may be configured to slidably receive a tether **60**. The rounded edge **198** may reduce stress on the tether **60** when the tether **60** exerts a force on the circular head fastener **180**.

[0303] The anti-rotation fastener **150** and/or the circular head fastener **180** may be configured within a range of lengths from 5 mm to 30 mm and within a range of diameters from 3 mm to 8 mm. The anti-rotation fastener **150** and/or the circular head fastener **180** may be one of a set of differently-sized implants, each having a different length and/or diameter.

[0304] The fracture plating system **100** may further include two or more locking nuts **30** and a two or more washers **50**. The locking nut **30** may be configured to threadably engage the anti-rotation fastener **150** and/or the circular head fastener **180** to secure the plate **102** to a portion of a rib. The locking nut **30** may be configured to receive the anti-rotation fastener **150** and/or the circular head fastener **180** and to cooperate with the anti-rotation fastener **150** and/or the circular head fastener **180** to secure the plate **102** to the bone. The washer **50** may be configured to be place between the locking nut **30** and a surface of a portion of a rib. The washer **50** may reduce damage to the surface of the portion of the rib due to rotation of the locking nut **30**. Additionally, or alternatively, the washer **50** may provide a greater contact area with the surface of the portion of the rib resulting in greater securing of the plate **102** with the portion of the rib.

[0305] FIG. **5**A is a perspective view of a locking nut **30** of the fracture plating system **100** according to an embodiment of the present disclosure. FIG. **5**B is a top view of the locking nut **30** and FIG. **5**C is a side view of the locking nut **30**. The locking nut **30** may include a threaded portion **32**, a slot **34**, an outside diameter **36**, and a thickness **38**. The threaded portion **32** may be configured to threadably engage the threaded portion **160** of the anti-rotation fastener **150** and/or the threaded portion **190** of the circular head fastener **180**. The slot **34** may allow engagement of a driver (not shown) to threadably engage the locking nut **30** with the anti-rotation fastener **150** and/or the circular head fastener **180**.

[0306] The outside diameter **36** may be larger than a hole sized to receive the anti-rotation fastener **150** and/or the circular head fastener **180**. The thickness **38** may be configured so that there is sufficient thread engagement to secure the plate **102** to the portion of the bone.

[0307] FIG. **6**A is a perspective view of a washer **50** of the fracture plating system **100** according to an embodiment of the present disclosure. FIG. **6**B is a top view of the washer **50** and FIG. **6**C is a side view of the washer **50**. The washer **50** may include an inside diameter **52**, an outside diameter **54**, and a thickness **56**. The inside diameter **52** may be configured to slidably receive the threaded portion **160** of the anti-rotation fastener **150** and/or the threaded portion **190** of the circular head fastener **180**. The outside diameter **54** may be configured to be greater than or equal to the outside diameter **36** of the locking nut **30**.

[0308] The thickness **56** may be configured so that when the washer **50** is placed between the surface of the portion of the rib and the locking nut **30**, a sufficient length of the threaded portion **160** of the anti-rotation fastener **150** and/or the threaded portion **190** of the circular head fastener **180** is exposed to allow sufficient thread engagement between the locking nut **30** and the anti-rotation fastener **150** and/or the circular head fastener **180** to secure the plate **102** to the portion of the rib.

[0309] FIG. **7** is a perspective view of the fracture plating system **100** in a lower profile configuration. The fracture plating system **100** may advantageously be configured so that, when the anti-rotation fastener **150** and/or the circular head fastener **180** are captive received within the first slot **105** and/or the second slot **115**, the anti-rotation fastener **150** and/or the circular head fastener **180** may be flattened against the plate **102** to make the construct lower profile and easier to insert into a patient.

[0310] FIG. **8** is a perspective view of the fracture plating system of **100** spanning an exemplary bone first fracture **21**. The fracture plating system **100** may include a second plate **102′**, securable to an exterior surface of a rib. The same fasteners may engage both a first plate **102** and a second plate **102′**.

[0311] FIG. **9** is a perspective view of the fracture plating system **100** in a lower profile configuration. FIG. **10**A is a perspective view of the plate **102** of the fracture plating system of **100** according to an embodiment of the present disclosure. FIG. **10**B is a top perspective partial view of the plate **102**. The first threaded feature **110** and the second threaded feature **120** may each be configured to threadably engage the threaded portion **190** to allow the threaded portion **190** to pass through the first threaded feature **110** and the second threaded feature **120** respectively. The circular head fastener **180** may be further configured so that, with the circular head fastener **180** captively received in the first slot **105** and/or the second slot **115**, the circular head fastener **180** may be moveable relative to the plate **102** along a longitudinal axis **182** of the circular head fastener **180**.

[0312] FIG. **11** is a front view of the fracture plating system **100** in a first pre-assembled configuration. The anti-rotation fastener **150** and/or the circular head fastener **180** may be aligned with the first threaded feature **110** and/or the second threaded feature **120** of the plate **102**.

[0313] FIG. **12** is a front view of the fracture plating system of **100** in a second pre-assembled configuration. The tip portion **162** of the anti-rotation fastener **150** and/or the tip portion **192** of the circular head fastener **180** may be slidably received within the first threaded feature **110** and/or the second threaded feature **120** of the plate **102**.

[0314] FIG. **13** is a front partial section view of the fracture plating system **100**. The threaded portion **160** of the anti-rotation fastener **150** and/or the threaded portion **190** of the circular head fastener **180** may threadably engage the first threaded feature **110** and/or the second threaded feature **120** of the plate **102**.

[0315] FIG. **14** is a front section view of the fracture plating system **100** in a third pre-assembled configuration. The head portion **155** of the anti-rotation fastener and/or the head portion **185** of the circular head fastener **180** may be configured to not pass through the first threaded feature **110** and/or the second threaded feature **120** of the plate **102** thereby capturing the anti-rotation fastener **150** and/or the circular head fastener **180** within the first slot **105** and/or the second slot **115** of the plate **102**.

[0316] FIG. **15** is a front view of the fracture plating system of **100** in an assembled configuration. FIG. **16** is a front section view of the fracture plating system **100** in an assembled configuration. In the assembled configuration, the fracture plating system **100** may include a plate assembly including a plate **102**, a first fastener, and a second fastener, wherein each of the first fastener and the second fastener may be chosen from the anti-rotation fastener and/or the head portion **185** of the circular head fastener **180**. The shank portion **157** of the anti-rotation fastener **150** and/or the shank portion **187** of the circular head fastener **180** may be configured to allow the anti-rotation fastener **150** and/or the circular head fastener **180** to translate along the first slot **105** and/or the second slot **115** of the plate **102**.

[0317] FIG. **17** is a front section view of the fracture plating system **100** in a lower profile configuration. The plate **102**, and the anti-rotation fastener **150** and/or the circular head fastener **180** may be configured so that the anti-rotation fastener **150** and/or the circular head fastener **180** may be flattened against the plate **102** to make the construct lower profile and easier to insert into an intra-thoracic cavity of a patient. The anti-rotation fastener **150** and/or the circular head fastener **180** may be configured to reside captive within the first slot **105** and/or the second slot **115** independently of engagement of any protruding feature of the head portion **155** and/or the head portion **185** with the plate **102**.

[0318] FIG. **18** is a perspective view of the fracture plating system **100** in a partially deployed configuration spanning an exemplary bone first fracture **21**. FIG. **19** is a front section view of the fracture plating system **100** in a partially deployed configuration spanning an exemplary bone first fracture **21**. The bone may include an interior surface **24** facing toward an interior body cavity of a patient, and an exterior surface **29** facing away from the interior body cavity of the patient.

[0319] The cannulation **165** of the anti-rotation fastener **150** and/or the cannulation **195** of the circular head fastener **180** may be configured to slidably receive a tether **60**. A tether **60** may be configured to apply a force directly to one or more fasteners to guide the one or more fasteners to the interior surface of the bone and through a hole in the interior surface of the bone.

[0320] A circular head fastener **180** length may be chosen from a range of fastener lengths such that the threaded portion **190** exposed beyond the exterior surface of the portion of bone may be greater than or equal to the thickness **38** of the locking nut **30**. Alternatively, a circular head fastener **180** length may be chosen from a range of fastener lengths such that the threaded portion **190** exposed beyond the exterior surface of the portion of bone may be greater than or equal to the thickness **38** of the locking nut **30** plus the thickness **56** of the washer **50**.

[0321] The tether **60** may be configured to guide the anti-rotation fastener **150** and/or the circular head fastener **180** through a first hole **15** and a second hole **16** from an interior side of a portion of a bone to an exterior side of a portion of a bone. The first hole **15** and the second hole **16** may be located on opposite sides of a first fracture **21**. The distance between the first hole **15** and the second hole **16** may be greater than the central portion **125** of the plate **102**.

[0322] FIG. **20** is a perspective view of the fracture plating system **100** in a partially deployed configuration spanning an exemplary bone first fracture **21**. FIG. **21** is a front section view of the fracture plating system **100** in a partially deployed configuration spanning an exemplary bone first fracture **21**. The tether **60** may be configured to pull the plate **102** against the interior side of a portion of a bone. The tether **60** may further be utilized to hold the plate **102** and the anti-rotation fastener **150** and/or the circular head fastener **180** in position so that the washer **50** and/or the locking nut **30** may be secured to the anti-rotation fastener **150** and/or the circular head fastener **180** to secure the fracture plating system **100** in place.

[0323] The fracture plating system **100** and the tether **60** may be configured so that a single tether **60** may be used to guide two fasteners through two holes in a portion of bone. The fracture plating system **100** and the tether **60** may further be configured so that a single tether **60** may be used to pull the plate **102** against an interior surface of a fractured rib.

[0324] FIG. **22** is a front section view of the fracture plating system **100** in a partially deployed configuration spanning an exemplary first fracture **21**. The tether **60** may include a first tether end **62** and a second tether end **64**. The first tether end **62** may be configured to be advanced through the cannulation **195** of a first circular head fastener **180** and the second tether end **64** may be configured to be advanced through the cannulation **195** of a second circular head fastener **180**.

[0325] FIG. **23** is a perspective view of the fracture plating system **100** in a partially deployed configuration spanning an exemplary first fracture **21**. The first tether end **62** may be pulled in the direction of the second circular head fastener **180** and the second tether end **64** may be pulled in the direction of the first circular head fastener, thereby applying compression to the first fracture **21** to facilitate reduction of the first fracture **21** by urging the first fastener and the second fastener towards each other. A locking nut **30** may be secured to each of the first and second circular head fastener **180** while compression is being applied to the fracture to secure the fracture plating system **100** in place.

[0326] FIG. **24** is a perspective view of the fracture plating system **100** in a partially deployed configuration spanning an exemplary bone first fracture **21**. FIG. **25** is a front view of the fracture plating system **100** in a partially deployed configuration spanning an exemplary bone first fracture **21**. The tether **60** may include a first bead **63** and a second bead **65**. The first bead **63** and the second bead **65** may be configured to engage the head portion **155** of the anti-rotation fastener **150** and/or the head portion **185** of the circular head fastener **180**. The first bead **63** and the second bead **65** may be configured as a solid spherical section of larger diameter.

[0327] Additionally, the first bead **63** and the second bead **65** may be larger than the cannulation **165** of the anti-rotation fastener **150** and/or the cannulation **195** of the circular head fastener **180**. The first bead **63** and the second bead **65** may not be received within the cannulation **165** of the anti-rotation fastener **150** and/or the cannulation **195** of the circular head fastener **180**.

[0328] The first bead **63**, the second bead **65**, and the anti-rotation fastener and/or the circular head fastener **180** may be configured so that, when the first bead **63** engages a first fastener, a tension force applied to the first tether end **62** is translated as a compression force applied to the first fastener. Similarly, when the second bead **65** engages a second fastener, a tension force applied to the second tether end **64** is translated as a compression force applied to the second fastener.

[0329] FIG. **26** is a perspective view of the fracture plating system **100** in a partially deployed configuration spanning an exemplary bone first fracture **21**. In an embodiment, the fracture plating system **100** may include a second plate **102′**. The second plate **102′** may be configured as an outer buttress plate. The second plate **102′** may be placed on an exterior surface of bone.

[0330] FIG. **27** is a perspective view of the fracture plating system **100** in a partially deployed configuration spanning an exemplary bone first fracture **21**. The second plate **102′** may span a first fracture **21** and may be secured to a portion of bone using the same fasteners that are used to secure the plate **102** to the interior surface of a bone.

[0331] FIG. **28** is a perspective view of the fracture plating system **100** in a deployed configuration spanning an exemplary bone first fracture **21**. The second plate **102′** may be secured to the exterior surface of bone with one or more locking nuts **30**. An anti-rotation fastener **150** length may be chosen from a range of fastener lengths such that, which the second plate **102′** proximate the exterior surface of the portion of bone, the threaded portion **160** exposed beyond the second plate **102′** may be greater than or equal to the thickness **38** of the locking nut **30**. Alternatively, an anti-rotation fastener **150** length may be chosen from a range of fastener lengths such that, which the second plate **102′** proximate the exterior surface of the portion of bone, the threaded portion **160** exposed beyond the second plate **102′** may be greater than or equal to the thickness **38** of the locking nut **30** plus the thickness **56** of the washer **50**.

[0332] FIG. **29** is a perspective view of a fracture plating system **200** according to an embodiment of the present disclosure. The fracture plating system **200** may include similar features as the fracture plating system **100** previously described. The fracture plating system **200** may include a fixed hinge fastener. The fixed hinge fastener may be configured to provide a lagging effect to facilitate compression of a fracture. The fracture plating system **200** may include a first fastener in a fixed location within a first slot of a plate and a second fastener that may be translated within a second slot of a plate.

[0333] The fracture plating system **200** may include a first fastener configured as one of a fixed hinge fastener **250** and a circular head fixed hinge fastener **280**. And a second fastener configured as one of an anti-rotation fastener **150** and a circular head fastener **180**. The first fastener may be captively received within a first slot **205** of a plate **202** so that the first fastener may be coupled to the first slot. The second fastener may be captively received within a second slot **215** of the plate **202** so that the second fastener may be coupled to the second slot.

[0334] The fracture plating system **200** may include a plate **202**. The plate **202** may include a plate radius **204**, a first slot **205**, a second slot **215**, a pin aperture **212**, and a central portion **225**. The first slot **205** may include a first threaded feature **210** and the second slot **215** may include a second threaded feature **220**. The central portion **225** may be generally in the center of the plate **202** and may separate the first slot **205** and the second slot **215**.

[0335] The first threaded feature **210** may be configured to threadably engage a threaded portion **260** of the fixed hinge fastener **250** and/or a threaded portion **290** of the circular head fixed hinge fastener **280** to allow the threaded portion **260** and/or the threaded portion **290** to pass through the first slot **205**. The second threaded feature **220** may be configured to threadably engage a threaded portion **160** of the anti-rotation fastener **150** and/or a threaded portion **190** of the circular head fastener **180** to allow the threaded portion **160** and/or the threaded portion **190** to pass through the second slot **215**.

[0336] The plate **202** may be configured such that a plate length is within a range of lengths from 30 mm to 300 mm. The plate **202** may be one of a set of differently-sized implants, each having a different plate length. The plate radius **204** may generally match a contour of an interior surface of a rib. The plate **202** may further be one of a set of differently-sized implants, each having a different plate radius **204**.

[0337] FIG. **30** is a perspective section view of the fracture plating system **200**. FIG. **31** is a partial bottom perspective view of the fracture plating system **200**. The pin aperture **212** may span the width of the plate **202** and may be located within the first slot **205**. The pin aperture **212** may further be configured to receive at least one hinge pin **240**. The pin aperture **212** may be configured to receive the at least one hinge pin **240** as a press fit so that the at least one hinge pin **240** may be held in place within the pin aperture **212**. Additionally, or alternatively, the at least one hinge pin **240** may be secured to the plate **202** within the pin aperture **212** by welding, solder, adhesive, or other suitable means.

[0338] FIG. **32**A is a top view of a circular head fixed hinge fastener **280** of the fracture plating system **200** according to an embodiment of the present disclosure. FIG. **32**B is a front view of the circular head fixed hinge fastener **280**. FIG. **32**C is a side view of the circular head fixed hinge fastener **280**. The circular head fixed hinge fastener **280** may include a head portion **285**, a shank portion **287**, a threaded portion **290**, a tip portion **292**, a cannulation **295**, and a pin aperture **299**.

[0339] The head portion **285** may be configured to be received within the first slot **205**. The threaded portion **290** may be larger than a width of the first slot **205** to prevent the circular head fixed hinge fastener **280** from disengaging from the plate **202**. The head portion **285** may include the pin aperture **299**. The pin aperture **299** may be configured to slidably receive at least one hinge pin **240**. The at least one hinge pin **240** may also be received within the pin aperture **212** and may prevent translation of the circular head fixed hinge fastener **280** within the first slot **205**.

[0340] The shank portion **287** may be smaller than a width of the first slot **205** to allow the circular head fixed hinge fastener **280** to translate along the length of the first slot **205**. The tip portion **292** may be configured to ease insertion of the circular head fixed hinge fastener **280** into a hole in a bone portion. The cannulation **295** may be configured to slidably receive a tether **60**.

[0341] The at least one hinge pin **240** may be configured to engage the pin aperture **212** of the plate **202** and a pin aperture **299** of the circular head fixed hinge fastener **280** so that the at least one hinge pin **240** is not received within the cannulation **295** and does not impede the tether **60** from passing through the cannulation **295**.

[0342] The circular head fixed hinge fastener **280** may be configured within a range of lengths from 5 mm to 30 mm and within a range of diameters from 3 mm to 8 mm. The circular head fixed hinge fastener **280** may be one of a set of differently-sized implants, each having a different length and/or diameter.

[0343] FIG. **33**A is a top view of a fixed hinge fastener **250** of the fracture plating system **200** according to an embodiment of the present disclosure. FIG. **33**B is a front view of the fixed hinge fastener **250**. FIG. **33**C is a side view of the fixed hinge fastener **250**. The fixed hinge fastener **250** may include a head portion **255**, a shank portion **257**, a threaded portion **260**, a tip portion **262**, a cannulation **265**, and a pin aperture **269**.

[0344] The head portion **255** may be configured to be received within the first slot **205**. The threaded portion **260** may be larger than a width of the first slot **205** to prevent the fixed hinge fastener **250** from disengaging from the plate **202**. The head portion **255** may include the pin aperture **269**. The pin aperture **269** may be configured to slidably receive at least one hinge pin **240**. The at least one hinge pin **240** may also be received within the pin aperture **212** and may prevent translation of the fixed hinge fastener **250** within the first slot **205**.

[0345] The shank portion **257** may be smaller than a width of the first slot **205** to allow the fixed hinge fastener **250** to translate along the length of the first slot **205**. The tip portion **262** may be configured to ease insertion of the fixed hinge fastener **250** into a hole in a bone portion. The cannulation **265** may be configured to slidably receive a tether **60**.

[0346] The at least one hinge pin **240** may be configured to engage the pin aperture **212** of the plate **202** and a pin aperture **269** of the fixed hinge fastener **250** so that the at least one hinge pin **240** is not received within the cannulation **265** and does not impede the tether **60** from passing through the cannulation **265**.

[0347] The fixed hinge fastener **250** may be configured within a range of lengths from 5 mm to 30 mm and within a range of diameters from 3 mm to 8 mm. The fixed hinge fastener **250** may be one of a set of differently-sized implants, each having a different length and/or diameter.

[0348] FIG. **34** is a perspective view of a fracture plating system **300** according to an embodiment of the present disclosure. The fracture plating system **300** may include similar features as the fracture plating system **100** and the fracture plating system **200** previously described. The fracture plating system **300** may include a fixed hinge fastener. The fixed hinge fastener may be configured to provide a lagging effect to facilitate compression of a fracture. The fracture plating system **300** may include a first fastener in a fixed location within a first slot of a plate and a second fastener that may be translated within a second slot of a plate.

[0349] The fracture plating system **300** may include a first fastener configured as a fixed hinge fastener **350**. And a second fastener configured an anti-rotation fastener **150**. The first fastener may be captively received within a first slot **305** of a plate **302**. The second fastener may be captively received within a second slot **315** of the plate **302**.

[0350] The fracture plating system **300** may include a plate **302**. The plate **302** may include a plate radius **304**, a first slot **305**, a second slot **315**, a pin channel **312**, and a central portion **325**. The first slot **305** may include a first threaded feature **310** and the second slot **315** may include a second threaded feature **320**. The central portion **325** may be generally in the center of the plate **302** and may separate the first slot **305** and the second slot **315**.

[0351] FIG. **35** is a bottom perspective view of the fracture plating system **300** in a pre-assembled configuration. The first threaded feature **310** may be configured to threadably engage a threaded portion **360** of the fixed hinge fastener **350** to allow the threaded portion **360** to pass through the first slot **305**. The second threaded feature **320** may be configured to threadably engage a threaded portion **160** of the anti-rotation fastener **150** and/or a threaded portion **190** of the circular head fastener **180** to allow the threaded portion **160** and/or the threaded portion **190** to pass through the second slot **315**.

[0352] The pin channel **312** may span the width of the plate **302** and may be located within the first slot **305**. The pin channel **312** may further be configured to receive at least one hinge pin **340**. The pin channel **312** may be configured to receive the at least one hinge pin **340** as a snap fit so that the at least one hinge pin **340** may be held in place within the pin channel **312**. The pin channel **312** may be configured so that a fixed hinge fastener **350** may be snapped into the plate **302** during a surgical procedure, and may allow a user to tailor the length of the fixed hinge fastener **350** based on an anatomy of a patient while still having the benefits of a fixed fastener for applying compression to a fracture. The fixed hinge fastener **350**, the plate **302**, and the pin channel **312** may be configured so that the, when the hinge pin **340** is snapped into the pin channel **312**, the fixed hinge fastener **350** may rotate about the hinge pin **340** but may not translate along the first slot **305**.

[0353] FIG. **36**A is a front view of a plate **302** of the fracture plating system **300** according to an embodiment of the present disclosure. FIG. **36**B is a bottom view of the plate **302**. The plate **302** may be configured such that a plate length is within a range of lengths from 30 mm to 300 mm. The plate **302** may be one of a set of differently-sized implants, each having a different plate length. A plate radius **304** may generally match a contour of an interior surface of a rib. The plate **302** may further be one of a set of differently-sized implants, each having a different plate radius **304**.

[0354] FIG. **37**A is a front view of a fixed hinge fastener **350** of the fracture plating system **300** according to an embodiment of the present disclosure. FIG. **37**B is a side view of the fixed hinge fastener **350**. FIG. **38**A is a front view of the anti-rotation fastener **150** and FIG. **38**B is a side view of the anti-rotation fastener **150**. The fixed hinge fastener **350** may include a head portion **355**, a shank portion **357**, a threaded portion **360**, a tip portion **362**, a cannulation **365**, and a pin aperture **369**.

[0355] The head portion **355** may be configured to be received within the first slot **305**. The threaded portion **360** may be larger than a width of the first slot **305** to prevent the fixed hinge fastener **350** from disengaging from the plate **302**. The head portion **355** may include the pin aperture **369**. The pin aperture **369** may be configured to slidably receive at least one hinge pin **340**. The at least one hinge pin **340** may also be received within the pin channel **312** and may prevent translation of the fixed hinge fastener **350** within the first slot **305**.

[0356] The shank portion **357** may be smaller than a width of the first slot **305** to allow the fixed hinge fastener **350** to translate along the length of the first slot **305**. The tip portion **362** may be configured to ease insertion of the fixed hinge fastener **350** into a hole in a bone portion. The cannulation **365** may be configured to slidably receive a tether **60**.

[0357] The at least one hinge pin **340** may be configured to engage the pin channel **312** of the plate **302** and a pin aperture **369** of the fixed hinge fastener **350** so that the at least one hinge pin **340** is not received within the cannulation **365** and does not impede the tether **60** from passing through the cannulation **365**. The at least one hinge pin **340** may be fabricated as a separate component from the fixed hinge fastener **350**. Alternatively, the at least one hinge pin **340** and the fixed hinge fastener **350** may be fabricated as a single integral component.

[0358] The fixed hinge fastener **350** may be configured within a range of lengths from 5 mm to 30 mm and within a range of diameters from 3 mm to 8 mm. The fixed hinge fastener **350** may be one of a set of differently-sized implants, each having a different length and/or diameter.

[0359] FIG. **39** is a perspective view of a fracture plating system **400** according to an embodiment of the present disclosure. The fracture plating system **400** may include similar features as the fracture plating system **100**, the fracture plating system **200**, and the fracture plating system **300** previously described. The fracture plating system **400** may include a ball-headed fastener **450**. The ball-headed fastener **450** may be configured to provide a lagging effect and/or poly-axial rotation of the ball-headed fastener **450** to facilitate compression of a fracture. The fracture plating system **400** may include a first fastener that may be translated within a first slot of a plate and a second fastener in a fixed location within a first slot of a plate.

[0360] FIG. **40** is a bottom perspective view of the fracture plating system **400**. The fracture plating system **400** may include a first fastener configured as a circular head fastener **180** and a second fastener configured as a ball-headed fastener **450**. The first fastener may be captively received within a first slot **405** of a plate **402**. The second fastener may be captively received within a second slot **415** of the plate **402**.

[0361] FIG. **41**A is a front view of a plate **402** of the fracture plating system **400** according to an embodiment of the present disclosure. FIG. **41**B is a bottom view of the plate **402**. The plate **402** may include a plate radius **404**, a first slot **405**, a second slot **415**, a socket **422**, and a central portion **425**. The first slot **405** may include a first threaded feature **410** and the second slot **415** may include a socket **422**. The central portion **425** may be generally in the center of the plate **402** and may separate the first slot **405** and the second slot **415**.

[0362] The first threaded feature **410** may be configured to threadably engage a threaded portion **190** of the circular head fastener **180** to allow the threaded portion **190** to pass through the first slot **305**.

[0363] The socket **422** may be located within the second slot **415**. The socket **422** may further be configured to receive a head portion **455** of the ball-headed fastener **450**. The socket **422** may be configured to receive the head portion **455** as a snap fit so that the head portion **455** may be held in place within the socket **422**. The socket **422** may be configured so that a ball-headed fastener **450** may be snapped into the plate **402** during a surgical procedure, and may allow a user to tailor the length of the ball-headed fastener **450** based on an anatomy of a patient while still having the benefits of a fixed fastener for applying compression to a fracture. The ball-headed fastener **450**, the plate **402**, and the socket **422** may be configured so that the, when the head portion **455** is snapped into the socket **422**, the ball-headed fastener **450** may rotate about the head portion **455** but may not translate along the second slot **415**.

[0364] The plate **402** may be configured such that a plate length is within a range of lengths from 30 mm to 300 mm. The plate **402** may be one of a set of differently-sized implants, each having a different plate length. A plate radius **404** may generally match a contour of an interior surface of a rib. The plate **402** may further be one of a set of differently-sized implants, each having a different plate radius **404**.

[0365] FIG. **42**A is a front view of the circular head fastener **180** and FIG. **42**B is a side view of the circular head fastener **180**. FIG. **43**A is a front view of a ball-headed fastener **450** of the fracture plating system **400** according to an embodiment of the present disclosure. FIG. **43**B is a side view of the ball-headed fastener **450**. The ball-headed fastener **450** may include a head portion **455**, a shank portion **457**, a threaded portion **460**, a tip portion **462**, and a cannulation **465**.

[0366] The head portion **455** may be configured to be received within the socket **422** and may prevent translation of the ball-headed fastener **450** within the second slot **415**. The shank portion **457** may be smaller than a width of the second slot **415** to allow the ball-headed fastener **450** to rotate about the head portion **455**. The tip portion **462** may be configured to ease insertion of the ball-headed fastener **450** into a hole in a bone portion. The cannulation **465** may be configured to slidably receive a tether **60**.

[0367] The ball-headed fastener **450** may be configured within a range of lengths from 5 mm to 30 mm and within a range of diameters from 3 mm to 8 mm. The ball-headed fastener **450** may be one of a set of differently-sized implants, each having a different length and/or diameter.

[0368] FIG. **44**A is a perspective view of a fracture plating system **500** according to an embodiment of the present disclosure. FIG. **44**B is a partial perspective view of the fracture plating system **500**. The fracture plating system **500** may include similar features as other fracture plating systems previously described within the present disclosure. The fracture plating system **500** may include a pin fastener. The pin fastener may be configured to translate along a length of a slot in a plate to facilitate compression of a fracture. The fracture plating system **500** may include a first fastener and a second fastener that each may be translated within a slot of a plate.

[0369] FIG. **45** is a bottom perspective view of the fracture plating system **500**. The fracture plating system **500** may include a first fastener configured as a pin fastener **550**. And a second fastener configured as a pin fastener **550**. The first fastener and the second fastener may be captively received within a first slot **505** and a second slot **515** of a plate **502**, respectively.

[0370] FIG. **46**A is a top view of a plate **502** of the fracture plating system **500** according to an embodiment of the present disclosure. FIG. **46**B is a front view of the plate **502**. The plate **502** may include a plate radius **504**, a first slot **505**, a second slot **515**, and a central portion **525**. The first slot **505** may include a first pocket **510** and the second slot **515** may include a second pocket **520**. The central portion **225** may be generally in the center of the plate **202** and may separate the first slot **205** and the second slot **215**.

[0371] The first pocket **510** and the second pocket **520** may be configured to receive a pin **540** received within a pin fastener **550**. The plate **502** may be one of a set of differently-sized implants, each having a different plate length. A plate radius **504** may generally match a contour of an interior surface of a rib. The plate **502** may further be one of a set of differently-sized implants, each having a different plate radius **504**.

[0372] FIG. **47**A is a top view of a pin fastener **550** of the fracture plating system **500** according to an embodiment of the present disclosure. FIG. **47**B is a front view of the pin fastener **550** and FIG. **47**C is a side view of the pin fastener **550**. The pin fastener may include a tab portion **555**, a shank portion **557**, a threaded portion **560**, a tip portion **562**, a cannulation **565**, and a pin aperture **570**.

[0373] The pin aperture **570** may be configured to securably receive at least one pin **540** such that, when the pin fastener **550** is assembled with the plate **502**, the at least one pin **540** is received within one of the first pocket **510** and the second pocket **520**. The tab portion **555** may be configured so that, when the at least one pin **540** is received with one of the first pocket **510** and the second pocket **520** and is aligned generally perpendicular to a long axis of the plate **502**, the tab portion **555** engages a bottom surface of the plate **502** to hold the pin fastener **550** captively received within one of the first slot **505** and the second slot **515** while allowing the pin fastener **550** to translate along a length of one of the first slot **505** and the second slot **515**.

[0374] The shank portion **557** may be smaller than a width of the first slot **505** and/or the second slot **515** to allow the pin fastener **550** to translate along the length of the first slot **505** and/or the second slot **515**. The tip portion **562** may be configured to ease insertion of the pin fastener **550** into a hole in a bone portion. The cannulation **565** may be configured to slidably receive a tether **60**. The threaded portion **560** may be configured to threadably receive a locking nut **30**.

[0375] The at least one pin **540** may be configured to be received within the first pocket **510** and/or the second pocket **520** of the plate **502** and a pin aperture **570** of the pin fastener **550** so that the at least one pin **540** is not received within the cannulation **565** and does not impede the tether **60** from passing through the cannulation **565**.

[0376] The pin fastener **550** may be configured within a range of lengths from 5 mm to 30 mm and within a range of diameters from 3 mm to 8 mm. The pin fastener **550** may be one of a set of differently-sized implants, each having a different length and/or diameter.

[0377] Each of the previously described fracture plating systems of the present disclosure may be utilized to treat multiple fractures of a single portion of bone. Each of the previously described fracture plating systems of the present disclosure may be assemblable in a modular fashion whereby a plate may be selectable from a range of differently sized plates, a plurality of fasteners may be selectable from a range of differently sized fasteners, and a locking nut may be selectable from a range of differently sized locking nuts. The fracture plating system may be configured to be assembled prior to implantation within a patient. The fracture plating system may be configured to be assembled on a back table during a surgical procedure prior to implantation within a patient. The fracture plating system may be configured so that a plate, two or more fasteners, and two or more locking nuts may be selected based on patient anatomy and/or the location and/or severity of one or more fractures. Additionally, or alternatively, the fracture plating system may be configured so that a plate, two or more fasteners, and two or more locking nuts may be selected based surgical and/or radiographic assessment of the patient and/or the location and/or severity of the one or more fractures.

[0378] FIG. **48**A is a front view of a fracture plating system **100** in a partially deployed configuration according to an embodiment of the present disclosure spanning a plurality of exemplary bone fractures. FIG. **48**B is a perspective view of a fracture plating system **100** secured to a bone with multiple fractures according to an embodiment of the present disclosure. FIG. **48**C is a perspective view of the fracture plating system **100** of FIG. **48**B. The fracture plating system **100** may include a plate **102′**, fasteners **150′**, and locking nuts **30′**.

[0379] The fracture plating system **100** may be configured to stabilize a multiple fractures of a bone of a patient, wherein the multiple fractures define one or more flail segments. For example, the fracture plating system **100** may be configured to stabilize a first fracture **21**, a second fracture **22** and a third fracture **23** of a bone, wherein the first fracture **21** and the second fracture **22** define a first flail segment **26′** and the second fracture **22** and the third fracture **23** define a second flail segment **27′**.

[0380] The fracture plating system may include a plate **102′** having a first slot **105′** and a second slot **115′**. The plate **102′** may span the first fracture **21**, the second fracture **22**, and the third fracture **23** so that at least one fastener **150′** may be received in the first slot **105′** or the second slot **115′** and in the bone on both sides of each of the first fracture **21**, the second fracture **22**, and the third fracture **23**. The first slot **105′** may span one or more of the first fracture **21**, the second fracture **22**, and the third fracture **23**. Additionally, the second slot **115′** may also span one or more of the first fracture **21**, the second fracture **22**, and the third fracture **23**.

[0381] FIG. **48**D is a perspective view of a fracture plating system **100** secured to a bone with multiple fractures according to an embodiment of the present disclosure. FIG. **48**E is a perspective view of the fracture plating system **100** of FIG. **48**D. The fracture plating system **100** may include a plate **102″**, fasteners **150″**, and locking nuts **30″**.

[0382] The fracture plating system **100** may be configured to stabilize a multiple fractures of a bone of a patient, wherein the multiple fractures define one or more flail segments. For example, the fracture plating system **100** may be configured to stabilize a first fracture **21**, a second fracture **22** and a third fracture **23** of a bone, wherein the first fracture **21** and the second fracture **22** define a first flail segment **26′** and the second fracture **22** and the third fracture **23** define a second flail segment **27′**.

[0383] The fracture plating system may include a plate **102″** having a first slot **105″**, a second slot **115″**, and a third slot **135″**. The plate **102″** may span the first fracture **21**, the second fracture **22**, and the third fracture **23** so that at least one fastener **150″** may be received in the first slot **105″**, the second slot **115′**, or the third slot **135″** and in the bone on both sides of each of the first fracture **21**, the second fracture **22**, and the third fracture **23**. The first slot **105″** may span one or more of the first fracture **21**, the second fracture **22**, and the third fracture **23**. Additionally, the second slot **115″** may also span one or more of the first fracture **21**, the second fracture **22**, and the third fracture **23**. Additionally, the third slot **135″** may also span one or more of the first fracture **21**, the second fracture **22**, and the third fracture **23**.

[0384] FIG. **49** is a front view of a fracture plating system in a partially deployed configuration according to an embodiment of the present disclosure spanning a plurality of exemplary bone fractures. FIG. **50** is a bottom perspective view of the fracture plating system in a partially deployed configuration spanning a plurality of exemplary bone fractures.

[0385] The fracture plating system may be configured to stabilize a single fracture of a bone. Additionally, or alternatively, the fracture plating system may be configured to stabilize two fractures of a bone. Additionally, or alternatively, the fracture plating system may be configured to stabilize three or more fractures of a bone. Additionally, or alternatively, the fracture plating system may be configured to stabilize multiple fractures of a bone, wherein the multiple fractures define one or more flail segments. The fracture plating system may include multiple tethers, each configured to guide up to two fasteners to an interior surface of the bone and to one or more slots of the plate.

[0386] The fracture plating system previously described within the present disclosure may be used to stabilize multiple fractures of a bone. The multiple fractures may result in one or more flail segments and may result in flail chest. The fracture plating system previously described within the present disclosure may be used to stabilize one or more flail segments.

[0387] The fracture plating system previously described within the present disclosure may include a plate that may be configured to span the multiple fractures and four fasteners, each configured to be received in the bone and in one or more slots of the plate. The four fasteners may be configured to be received in the bone on opposite sides of each of the first fracture **21** and the second fracture **22**. The fracture plating system may also include four locking nuts configured to receive one of the four fasteners and cooperate with one of the four fasteners to secure the plate to the interior surface of the bone.

[0388] Additionally, or alternatively, the fracture plating system previously described within the present disclosure may include a plate that may be configured to span a first fracture **21** and a second fracture **22** and at least three fasteners, each configured to be received in the bone and in one or more slots of the plate. The at least three fasteners may be configured to be received in the bone on opposite sides of each of the first fracture **21** and the second fracture **22**. The fracture plating system may also include at least three locking nuts configured to receive one of the at least three fasteners and cooperate with one of the at least three fasteners to secure the plate to the interior surface of the bone.

[0389] The fracture plating system previously described within the present disclosure may be used to facilitate reduction of a first fracture **21**, a second fracture **22**, and a third fracture **23** whereby a first tether **60** may be received within a first fastener that is received within a first bone portion **25** and a fourth fastener that may be received within a fourth bone portion **28**. A second tether **60** may be received within a second fastener that may be received within a second bone portion **26** and a third fastener that may be received within a third bone portion **27**.

[0390] Applying a tension force to a first tether end **62** and a second tether end **64** of a first tether **60** and a first tether end **62** and a second tether end **64** of a second tether **60** may facilitate reduction of the first fracture **21**, the second fracture **22**, and the third fracture **23**. The fracture plating system may then be secured by securing a locking nut to each of the fasteners.

[0391] The fracture plating system previously described within the present disclosure may be used to facilitate reduction of a first fracture **21** and a second fracture **22** whereby a first tether **60** is received within a first fastener that may be received within a first bone portion **25** and a second fastener that may be received within a second bone portion **26**. A second tether **60** may be received within a third fastener that may be received within a third bone portion **27** and a fourth fastener that may be received within a fourth bone portion **28**.

[0392] Applying a tension force to a first tether end **62** and a second tether end **64** of a first tether **60** and a first tether end **62** and a second tether end **64** of a second tether **60** may facilitate reduction of the first fracture **21** and the second fracture **22**. The fracture plating system may then be secured by securing a locking nut to each of the fasteners.

[0393] FIG. **51** is a bottom perspective view of a fracture plating system in a partially deployed configuration according to an embodiment of the present disclosure spanning an exemplary bone fracture. FIG. **52** is a partial bottom perspective view of the fracture plating system of FIG. **51** spanning an exemplary bone fracture. A tether **60** may be configured so that a first tether end **62** may releasably connect to a second tether end **64**. The tether **60** may fed through two fasteners and a plate in the opposite direction, i.e.: the first tether end **62** passing through a tip portion of a first fastener, the second tether end **64** passing through a tip portion of a second fastener, and the first tether end **62** releasably connecting to the second tether end **64** on the bottom side of the plate. The first tether end **62** and the second tether end **64** tether ends may then be releasably connected together to create a single continuous loop of tether. The first tether end **62** and the second tether end **64** may each include a connection feature **68**.

[0394] A first hole **15** may be drilled in a first bone portion **25** and a second hole **16** may be drilled in a second bone portion **26**. The first tether end **62** may be passed through the first hole **15** and the second tether end **64** may be passed through the second hole **16**. The first tether end **62** and the second tether end **64** may then be grabbed with an endoscopic grasper and pulled out thru a VATS port. The first tether end **62** may then be passed through the first fastener and a first slot of a plate and the second tether end **64** may then be passed through the second fastener and a second slot of the plate. The first tether end **62** may then be connected to the second tether end **64** using the connection feature **68**. Once the connection is made, the tether **60** may then be pulled to place the plate again the interior surface of a portion of bone.
[0395] FIG. **53** is a front view of a fracture plating system in a partially deployed configuration according to an embodiment of the present disclosure spanning an exemplary bone fracture. FIG. **54** is a partial front view of the fracture plating system of FIG. **53** spanning an exemplary bone fracture. FIG. **55** is a partial bottom perspective view of the fracture plating system of FIG. **53** spanning an exemplary bone fracture.
[0396] A tether **60** may be configured so that a first tether end **62** and a second tether end **64** may each include a toggle feature **69**. The tether **60** may fed through two fasteners and a plate in the opposite direction, i.e.: the first tether end **62** passing through a tip portion of a first fastener, the second tether end **64** passing through a tip portion of a second fastener, a first toggle feature **69** may be deployed so that the first tether end **62** may not return through the first fastener, and a second toggle feature may be deployed so that the second tether end **64** may not return through the second fastener.
[0397] A first hole **15** may be drilled in a first bone portion **25** and a second hole **16** may be drilled in a second bone portion **26**. The first tether end **62** may be passed through the first hole **15** and the second tether end **64** may be passed through the second hole **16**. The first tether end **62** and the second tether end **64** may then be grabbed with an endoscopic grasper and pulled out thru a VATS port. The first tether end **62** may then be passed through the first fastener and a first slot of a plate and the second tether end **64** may then be passed through the second fastener and a second slot of the plate. The toggle feature **69** of the first tether end **62** and the toggle feature **69** of the second tether end **64** may then be deployed. The tether **60** may then then be pulled to place the plate again the interior surface of a portion of bone.
[0398] FIG. **56** is a front view of a fracture plating system **600** according to an embodiment of the present disclosure spanning a plurality of exemplary bone fractures. FIG. **57** is a bottom perspective view of the fracture plating system **600** spanning a plurality of exemplary bone fractures. The fracture plating system **600** may include similar features as other fracture plating systems previously described within the present disclosure. The fracture plating system **600** may include a plate **602** and may include two or more of an anti-rotation fastener **150** and/or a circular head fastener **180**.
[0399] FIG. **58**A is a front view of a plate **602** of the fracture plating system **600** according to an embodiment of the present disclosure. FIG. **58**B is a bottom view of the plate **602**. The plate **602** may include a plate radius **604**, a slot **605**, a first threaded feature **610**, and a second threaded feature **620**. Alternatively, the plate **602** may include a plate radius **604**, a slot **605**, and a first threaded feature **610** whereby a plurality of fasteners may threadably engage the first threaded feature **610** to allow the plurality of fasteners to be slidably received within the slot **605**.
[0400] The plate **602** may have one continuous slot **605** that allows the user to choose the number of fasteners to be added to the fracture plating system **600** and may allow for more intraoperative flexibility of where a fastener may be located relative to the plate **602** and relative to a fracture. The number of fasteners may be customized to each patient to match their individual facture pattern. Similar to fracture plating system previously described in the present disclosure, the fasteners may be connected to the plate **602** during a surgical procedure by threading the two or more fasteners through the first threaded feature **610** and/or the second threaded feature **620** of the plate **602**.

Multiple lengths of fasteners may be selected and attached to the plate **602** to match the thickness of the rib in that region. Additional points of fixation (a 3 or more fasteners as shown) may accommodate for variations in curvature and/or thickness of a portion of bone.
[0401] The plate **602** may be one of a set of differently-sized implants, each having a different plate length. A plate radius **604** may generally match a contour of an interior surface of a rib. The plate **602** may further be one of a set of differently-sized implants, each having a different plate radius **604**.
[0402] FIG. **59** is a perspective view of a fracture plating system **700** according to an embodiment of the present disclosure spanning a plurality of exemplary bone fractures. FIG. **60** is a perspective view of the fracture plating system **700** spanning a plurality of exemplary bone fractures. The fracture plating system **700** may include similar features as other fracture plating systems previously described within the present disclosure. The fracture plating system **700** may include a plate **702**, a threaded fastener **750**, and an anti-rotation fastener **150** and/or a circular head fastener **180**. The fracture plating system **700** may allow a user to use a single plate **702** and a single tether **60** to stabilize two fractures in a single portion of bone.
[0403] FIG. **61** is a bottom perspective view of the fracture plating system **700** spanning a plurality of exemplary bone fractures. The plate **702** may be secured to a first bone portion **25** and a third portion **27** using components and methods previously described. The user may then drill a hole through the second bone portion **26** and through a central portion **725** of the plate **702** creating a central aperture **730** in the plate **702**. A threaded fastener **750** may then be threadably inserted through the drilled hole and into central aperture **730** to secure the second bone portion to the plate **702**.
[0404] FIG. **62**A is a front view of a plate **702** of the fracture plating system **700** according to an embodiment of the present disclosure. FIG. **62**B is bottom view of the plate **702**. The plate **702** may include a plate radius **704**, a first slot **705**, a second slot **715**, and a central portion **725**. The first slot **705** may include a first threaded feature **710** and the second slot **715** may include a second threaded feature **720**. The central portion **725** may be generally in the center of the plate **702** and may separate the first slot **705** and the second slot **715**.
[0405] The first threaded feature **710** may be configured to threadably engage a threaded portion **190** of a circular head fastener **180** and/or the threaded portion **160** of an anti-rotation fastener **150** to allow the threaded portion **190** and/or the threaded portion **160** to pass through the first slot **705**. The second threaded feature **720** may be configured to threadably engage a threaded portion **190** of a circular head fastener **180** and/or the threaded portion **160** of an anti-rotation fastener **150** to allow the threaded portion **190** and/or the threaded portion **160** to pass through the second slot **715**.
[0406] The plate **702** may be configured such that a plate length is within a range of lengths from 30 mm to 300 mm. The plate **702** may be one of a set of differently-sized implants, each having a different plate length. A plate radius **704** may generally match a contour of an interior surface of a rib. The plate **702** may further be one of a set of differently-sized implants, each having a different plate radius **704**.
[0407] FIG. **63**A is a front view of a threaded fastener **750** of the fracture plating system **700** according to an embodiment of the present disclosure. FIG. **63**B is a side view of the threaded fastener **750**. The threaded fastener **750** may include a head portion **755**, a shank portion **757**, a threaded portion **760**, a tip portion **762**, a cannulation **765**, and a drive portion **770**.
[0408] The head portion **755** may include the drive portion **770**. The drive portion may be configured to receive a driver instrument (not shown) configured to impart a torque to the threaded fastener **750**. The drive portion **770** may be configured as a hex, a hexalobe, a square, a slot, or other drive geometry known in the art.
[0409] The cannulation **765** may be configured to receive a guide wire (not shown) to assist in placement location of the threaded fastener **750**. The threaded portion **760** and the shank portion **757** may be configured to receive a washer **50**. The tip portion **762** may be configured to guide the

threaded fastener **750** into a hole drilled into a portion of bone. The threaded portion **760** may be configured to threadably engage the second bone portion **26** and the central aperture **730**. Alternatively, a diameter of the hole drilled onto a portion bone may be larger than a thread diameter of the threaded fastener **750** and the threaded portion **760** may be configured to threadably engage the central aperture **730**.

[0410] The threaded fastener **750** may be configured within a range of lengths from 5 mm to 20 mm and within a range of diameters from 3 mm to 8 mm. The threaded fastener **750** may be one of a set of differently-sized implants, each having a different length and/or diameter.

[0411] FIG. **64**A front view of a toggle fastener **850** in an insertion configuration of a fracture plating system **800** according to an embodiment of the present disclosure. FIG. **64**B is a front view of the toggle fastener **850** in a deployed configuration. The fracture plating system **800** may include similar features as other fracture plating systems previously described within the present disclosure. The fracture plating system **800** may include at least one toggle fastener **850**, at least one locking nut **30** and a plate.

[0412] In an embodiment, at least one toggle fastener **850** may be inserted thru at least one hole in a bone. The at least one toggle fastener **850** may then be grabbed with an endoscopic grasper and pulled out a VATS port. The at least one toggle fastener **850** may then be inserted thru at least one slot in a properly configured plate and the at least one toggle fastener **850** may then be deployed to engage the plate. At least one tether **60** tethers may be pre-attached to a top of the toggle fastener. A user may pull the at least one tether **60** to tension the plate against the bone.

[0413] The toggle fastener **850** may include a head portion **855**, a shank portion **857**, a threaded portion **860**, a tip portion **862**, at least one toggle **880**, and a pin **890**. The head portion **855** may be configured to guide a washer **50** and/or a locking nut **30** onto the toggle fastener **850**. The shank portion **857** may be configured to receive at least on toggle **880** when the at least one toggle is in an insertion configuration. The threaded portion **860** may be configured to threadably reactive a locking nut **30**. The pin may be configured to rotatably secure at least one toggle **880** to the shank portion **857**. The at least one toggle **880** may be configured so that, in a deployed configuration, the at least one toggle **880** extends beyond the threaded portion **860**.

[0414] FIG. **65** is a perspective view of a pair of toggle fasteners **850** in a partially deployed configuration spanning an exemplary bone fracture. FIG. **66** is a perspective view of a pair of toggle fasteners **850** in a partially deployed configuration spanning an exemplary bone fracture. FIG. **67** is a perspective view of a pair of toggle fasteners **850** in a partially deployed configuration spanning an exemplary bone fracture. A first hole **15** may be drilled in a first bone portion **25** and a second hole **16** may be drilled in a second bone portion **26**.

[0415] FIG. **68** is a perspective view of a pair of toggle fasteners **850** in a partially deployed configuration spanning an exemplary bone fracture. The toggle fastener **850** may include an insertion configuration in which a first toggle **880** and a second toggle **880** may be aligned generally parallel with a long axis of the toggle fastener **850**. The toggle fastener **850** may include a deployed configuration in which a first toggle **880** and a second toggle **880** may be generally perpendicular to a long axis of the toggle fastener **850**.

[0416] FIG. **69** is a perspective view of a fracture plating system **800** including the toggle fastener **850** according to an embodiment of the present disclosure spanning an exemplary bone fracture. A first toggle fastener **850**, in an insertion configuration, may be inserted through the first hole, until the first toggle **880** and the second toggle **880** are past the inner surface of the bone. The first toggle fastener **850** may then translate from the insertion configuration to the deployed configuration and the toggle fastener **850** may be drawn upward and the toggle fastener **850** contacts the inside surface of the bone.

[0417] FIG. **70** is a bottom perspective view of the fracture plating system **800** spanning an exemplary bone fracture. The steps may be repeated for a second toggle fastener **850** being inserted into a second hole **16**. The first toggle fastener **850** and the second toggle fastener **850** may be

urged towards each other to reduce the facture.
[0418] FIG. **71** is a bottom view of the plate **102**. A plate **102** may be applied to the tops of the first toggle fastener **850** and the second toggle fastener **850** fastener to provide rigid fixation that bridges the gap of the fracture. A washer **50** a locking nut **30** may be secured to each of the first toggle fastener **850** and the second toggle fastener **850** to tighten the construct in place.
[0419] FIG. **72** is a perspective view of a fracture plating system **900** according to an embodiment of the present disclosure in a partially assembled configuration. FIG. **73** is a perspective view of the fracture plating system **900** in a partially assembled configuration. The fracture plating system **900** may include similar features as other fracture plating systems previously described within the present disclosure.
[0420] The fracture plating system **900** may include a post fastener **950**, a locking nut **30**, and one of any of the plates previously described, for example, plate **102**. The post fastener **950** may be received in a first slot **105** and/or a second slot **115** of the plate **102**. A nut **970** may then be threadably engaged with a second threaded portion **955** of the post fastener **950** to hold the post fastener **950** captively received within the plate **102** while still allowing translation of the post fastener **950** along a longitudinal axis of a first slot **105** and/or a second slot **115**.
[0421] FIG. **74**A is a front view of a post fastener **950** of the fracture plating system **900** according to an embodiment of the present disclosure. FIG. **74**B is a side view of the post fastener **950**. The post fastener **950** may include a first threaded portion **960**, a second threaded portion **955**, a shank portion **957**, a tip portion **962**, and a cannulation **965**.
[0422] The shank portion **957** may be smaller than a width of the first slot **105** and/or the second slot **115** to allow the post fastener **950** to translate along the length of the first slot **105** and/or the second slot **115**. The tip portion **962** may be configured to ease insertion of the post fastener **950** into a hole in a bone portion. The cannulation **965** may be configured to slidably receive a tether **60**. The first threaded portion **960** may be configured to threadably receive a locking nut **30**. The second threaded portion **955** may be configured to receive a nut **970**.
[0423] FIG. **75**A is a front view of a nut **970** of the fracture plating system **900** according to an embodiment of the present disclosure. FIG. **75**B is a side view of the nut **970** and FIG. **75**C is a perspective view of the nut **970**. The nut **970** may include a third threaded portion **975**, a height **980**, and an outer portion **985**. The third threaded portion **975** may be configured to threadably engage the second threaded portion **955** to capture the post fastener **950** within the first slot **105** and/or the second slot **115** of the plate **102** while still allowing translation of the post fastener **950** along the first slot **105** and/or the second slot **115**.
[0424] The height **980** may be configured to maximize threaded engagement between the second threaded portion **955** and the third threaded portion **975**. The outer portion **985** may be configured as a not circular geometry to prevent rotation of the nut **970** when the nut **970** is received within the first slot **105** and/or the second slot **115**.
[0425] FIG. **76**A is top view of a pin fastener **1050** of a fracture plating system **1000** according to an embodiment of the present disclosure. FIG. **76**B is a perspective view of the pin fastener **1050**, FIG. **76**C is a front view of the pin fastener **1050**, and FIG. **76**D is a side view of the pin fastener **1050**.
[0426] The fracture plating system **1000** may include similar features as other fracture plating systems previously described within the present disclosure. The fracture plating system **1000** may include a pin fastener **1050**. The pin fastener **1050** may be configured to translate along a length of a first slot **1005** and/or a second slot **1015** in a plate **1002** to facilitate compression of a fracture. The fracture plating system **1000** may include a first pin fastener **1050** and a second pin fastener **1050** that each may translate within the first slot **1005** and/or the second slot **1015** of the plate **1002**.
[0427] The pin fastener **1050** may include a head portion **1053**, a threaded portion **1060**, a tip portion **1062**, and a cannulation **1065**. The head portion **1053** may include at least one pin portion

1055. The head portion **1053** may be configured to be received within the first slot **1005** and/or the second slot **1015**. The head portion **1053** may have a non-circular profile so that, when the head portion **1053** is received within the first slot **1005** and/or the second slot **1015**, the pin fastener **1050** is prevented form rotating around the long axis of the pin fastener **1050**, specifically when a locking nut **30** is threadably engaging the threaded portion **1060**.

[0428] The pin portion **1055** may be configured to be received within a first groove **1012** and/or a second groove **1022** of the plate **1002**. The pin portion **1055** may have a generally circular profile so that the pin fastener **1050** may translate along the length of the first slot **1005** and/or the second slot **1015** and the pin fastener **1050** may rotate about a long axis of the pin portion **1055**.

[0429] The tip portion **1062** may be configured to ease insertion of the pin fastener **1050** into a hole in a bone portion. The cannulation **1065** may be configured to slidably receive a tether **60**. The threaded portion **1060** may be configured to threadably receive a locking nut **30**.

[0430] The pin fastener **1050** may be configured within a range of lengths from 5 mm to 30 mm and within a range of diameters from 3 mm to 8 mm. The pin fastener **1050** may be one of a set of differently-sized implants, each having a different length and/or diameter.

[0431] FIG. **77**A is a top view of a plate **1002** of the fracture plating system **1000** according to an embodiment of the present disclosure. FIG. **77**B is a front view of the plate **1002**. The plate **1002** may include a plate radius **1004**, a first slot **1005**, a second slot **1015**, and a central portion **1025**. The first slot **1005** may include a first pocket **1010** and a first groove **1012**. The second slot may include a second pocket **1020** and a second groove **1022**.

[0432] FIG. **78** is a partial top view of the fracture plating system **1000** in a partially assembled configuration. FIG. **79** is a partial perspective view of the fracture plating system **1000** in a partially assembled configuration. FIG. **80** is a perspective section view of the fracture plating system **1000**. The central portion **1025** may be generally in the center of the plate **1002** and may separate the first slot **1005** and the second slot **1015**.

[0433] FIG. **81** is a perspective view of a fracture plating system **1000** according to an embodiment of the present disclosure. FIG. **82** is a partial perspective view of the fracture plating system **1000**. FIG. **83** is a partial perspective view of the fracture plating system **1000**.

[0434] The first pocket **1010** may be configured to receive the head portion **1053** and the pin portion **1055** of the pin fastener **1050**. After the pin portion **1055** is received within the first pocket **1010**, the pin portion **1055** may be slidably received into the first groove **1012**. The second pocket **1020** may be configured to receive the head portion **1053** and the pin portion **1055** of the pin fastener **1050**. After the pin portion **1055** is received within the second pocket **1020**, the pin portion **1055** may be slidably received into the second groove **1022**.

[0435] The plate **1002** may be one of a set of differently-sized implants, each having a different plate length. A plate radius **1004** may generally match a contour of an interior surface of a rib. The plate **1002** may further be one of a set of differently-sized implants, each having a different plate radius **1004**.

[0436] FIG. **102** is a perspective view of a fracture plating system **1100** according to an embodiment of the present disclosure spanning an exemplary bone fracture. FIG. **103** is a perspective view of a fracture plating system **1100**. The fracture plating system **900** may include similar features as other fracture plating systems previously described within the present disclosure.

[0437] The fracture plating system **1100** may include an anti-rotation fastener **1150**, a locking nut **1170**, and one of any of the plates previously described, for example, plate **102**. The anti-rotation fastener **1150** may be received in a first slot **105** and/or a second slot **115** of the plate **102**. A locking nut **1170** may then be threadably engaged with a threaded portion **1160** of the anti-rotation fastener **1150** to hold the anti-rotation fastener **1150** captively received within the plate **102** while still allowing translation of the anti-rotation fastener **1150** along a longitudinal axis of the first slot **105** and/or a second slot **115**. The anti-rotation fastener **1150** may include a head portion **1155**, a shank portion **1157**, a threaded portion **1160**, a tip portion **1162**, a cannulation **1165**, and a rounded edge **1168**.

[0438] FIG. **104**A is a front view of an anti-rotation fastener **1150** of the fracture plating system **1100** according to an embodiment of the present disclosure. FIG. **104**B is a side view of the anti-rotation fastener **1150**. The head portion **1155** may be configured to be received within the first slot **105** and/or the second slot **115**. The threaded portion **1160** may be larger than a width of the first slot **105** and/or the second slot **115** to prevent the anti-rotation fastener **1150** from disengaging from the plate **102**.

[0439] The shank portion **1157** may be smaller than a width of the first slot **105** and/or the second slot **115** to allow the anti-rotation fastener **1150** to translate along the length of the first slot **105** and/or the second slot **115**. The tip portion **1162** may be configured to ease insertion of the anti-rotation fastener **1150** into a hole in a bone portion. The cannulation **1165** may be configured to slidably receive a tether **60**. The threaded portion **1160** may be configured to threadably receive a locking nut **1170**. The rounded edge **1168** may reduce stress on the tether **60** when the tether **60** exerts a force on the anti-rotation fastener **1150**.

[0440] The anti-rotation fastener **1150** may be configured within a range of lengths from 5 mm to 30 mm and within a range of diameters from 2 mm to 8 mm. The anti-rotation fastener **1150** may be one of a set of differently-sized implants, each having a different length and/or diameter.

[0441] FIG. **105**A is a front view of a locking nut **1170** of the fracture plating system **1100** according to an embodiment of the present disclosure. FIG. **105**B is a side view of the locking nut **1170**. The locking nut **1170** may be configured to threadably engage the anti-rotation fastener **1150** to secure the plate **102** to a portion of a rib.

[0442] The locking nut **1170** may include a threaded portion **1172**, a body **1174**, a body diameter **1176**, a head profile **1178**, and a body length **1180**. The threaded portion **1172** may be configured to threadably engage the threaded portion **1160** of the anti-rotation fastener **1150**. The head profile **1178** may allow engagement of a driver **1200** to threadably engage the locking nut **1170** with the anti-rotation fastener **1150**. The head profile **1178** may be configured as a hex, a square, or other non-circular geometry.

[0443] The body diameter **1176** may be configured to be received in a hole drilled into a portion of bone. The body length **1180** may be configured to be less than a thickness of a portion of the bone.

[0444] FIG. **106**A is a perspective view of a driver **1200** of the fracture plating system **1100** according to an embodiment of the present disclosure. FIG. **106**B is a side view of the driver **1200** and FIG. **106**C is a front view of the driver **1200**. The driver **1200** may be configured to engage a locking nut **1170** and may transfer torque from a handle (not shown) to the locking nut **1170** thereby threadably securing the locking nut **1170** to an anti-rotation fastener **1150**.

[0445] The driver **1200** may include a quick connect feature **1210**, a head portion **1260**, and a shaft configured to connect the quick connect feature **1210** and the head portion **1260**. The shaft **1220** may include a cannulation **1250**. The head portion **1260** may include a socket portion **1230** that may include a drive portion **1240**. The cannulation **1250** may be configured to receive a guide wire (not shown) to assist in alignment of the driver **1200** with the anti-rotation fastener **1150**.

[0446] The quick connect feature **1210** may be configured to be removably received by a handle (not shown) having a compatible quick connect feature. The quick connect feature **1210** may be configured as one of: AO connector, Hudson connector, trilobe connector, square connector, hex connector, Stryker connector, Stryker-Hall connector, or other quick connect mechanism known in the art.

[0447] The socket portion **1230** may be configured to receive the head profile **1178** of the locking nut **1170**. The drive portion **1240** may be configured to engage the head profile **1178** and may be configured as a hex, a double hex, a square or other non-circular geometry that may engage the head profile **1178** of the locking nut **1170** in order to transfer torque from the driver **1200** to the locking nut **1170**.

[0448] FIG. **107** is a front perspective view of a spinal fixation plating system **1300**, secured to an exemplary portion of a spine, according to an embodiment of the present disclosure. The spinal fixation plating system **1300** may include a plate **1302**, an anti-rotation fastener **1350**, and a locking nut **1370**. The plate **1302** may span an intervertebral space **14** and may be secured to a first vertebra **10** and a second vertebra **12**.

[0449] The spinal fixation plating system **1300** may be configured to provide immobilization and stabilization of spinal segments. The spinal fixation plating system **1300** may be used as an adjunct to fusion in the treatment of the cervical, thoracic, lumbar, and/or sacral spine. Additionally, or alternatively, the spinal fixation plating system **1300** may be used in conjunction with other devices to support fusion of one or more spinal segments. The spinal fixation plating system **1300** may be implanted through an interior approach (ALIF), a posterior transforaminal approach (TLIF), a lateral approach (DLIF/XLIF), and/or an oblique approach (OLIF).

[0450] One or more spinal fixation plating system **1300** constructs may be used on a single spinal segment. The one or more spinal fixation plating system **1300** constructs may be implanted on opposite side of the spinal column. Additionally, or alternatively, a single spinal fixation plating system **1300** construct may span one or more spinal segments. Additionally, or alternatively, one or more spinal fixation plating system **1300** constructs may be configured to prevent expulsion and/or subsidence of one or more intervertebral implants. Additionally, or alternatively, the spinal fixation plating system **1300** may be configured to be implanted as an adjunct to a posterior rod/pedicle screw construct.

[0451] FIG. **108** is a rear perspective view of the spinal fixation plating system **1300** secured to an exemplary portion of a spine. Similar to the fracture plating systems previously described, the spinal fixation system may be configured to be deployed using a single tether **60** to pull the plate against the first vertebra **10** and the second vertebra **12**. The tether **60** may be used to guide a first anti-rotation fastener **1350** through a hole in the first vertebra **10** and a second anti-rotation fastener **1350** through a hole in the second vertebra **12**. A first locking nut **1370** may be guided along a first tether end **62** to threadably engage the first anti-rotation fastener **1350**. A second locking nut **1370** may be guided along a second tether end **64** to threadably engage the second anti-rotation fastener **1350**.

[0452] FIG. **109** is a front perspective view of the spinal fixation plating system **1300** secured to an exemplary portion of a spine. FIG. **110** is a rear perspective view of the spinal fixation plating system **1300** secured to an exemplary portion of a spine. After the first locking nut **1370** threadably engages the first anti-rotation fastener **1350** and the second locking nut **1370** threadably engages the second anti-rotation fastener, the tether **60** may be removed.

[0453] FIG. **111**A is a front view of a plate **1302** of the spinal fixation plating system **1300** according to an embodiment of the present disclosure. FIG. **111**B is a bottom view of the plate **1302**. The plate **1302** may include a plate length **1303** and a plate radius **1304**. The plate length **1303** may be configured to span one or more spinal segments.

[0454] The plate **1302** may be configured such that the plate length **1303** is within a range of lengths from 30 mm to 300 mm. The plate **1302** may be one of a set of differently-sized implants, each having a different plate length **1303**. The plate radius **1304** may generally match a contour of a lateral and/or anterior portion of a vertebra. The plate **1302** may further be one of a set of differently-sized implants, each having a different plate radius **1304**.

[0455] The plate **1302** may further include a central portion **1325** and a first slot **1305** having a first slot width **1307** and a first threaded feature **1310**. The plate **1302** may also include a second slot **1315** having a second slot width **1317** and a second threaded feature **1320**. The central portion **1325** may be generally in the center of the plate **1302** and may separate the first slot **1305** and the second slot **1315**.

[0456] The first slot width **1307** and the second slot width **1317** may each be configured to receive an anti-rotation fastener **1350**. The first slot width **1307** and the second slot width **1317** may further be configured to allow translation of the anti-rotation fastener **1350** along the length of the first slot

1305 and the second slot **1315** respectively.

[0457] FIG. **112**A is a perspective view of a locking nut **1370** of the spinal fixation plating system **1300** according to an embodiment of the present disclosure. FIG. **112**B is a front view of the locking nut **1370** and FIG. **112**C is a side view of the locking nut **1370**. The locking nut **1370** may be configured to threadably engage the anti-rotation fastener **1350** to secure the plate **1302** to a vertebra.

[0458] The locking nut **1370** may include a threaded portion **1372**, a body **1374**, a body diameter **1376**, a head profile **1378**, and a body length **1380**. The threaded portion **1372** may be configured to threadably engage the threaded portion **1360** of the anti-rotation fastener **1350**. The head profile **1378** may allow engagement of a driver **1200** to threadably engage the locking nut **1370** with the anti-rotation fastener **1350**. The head profile **1378** may be configured as a hex, a square, or other non-circular geometry.

[0459] The body diameter **1376** may be configured to be received in a hole drilled into a vertebra. The body length **1380** may be configured to be less than a thickness of a vertebra.

[0460] FIG. **113**A is a perspective view of an anti-rotation fastener **1350** of the spinal fixation plating system **1300** according to an embodiment of the present disclosure. FIG. **113**B is a front view of the anti-rotation fastener **1350** and FIG. **113**C is a side view of the anti-rotation fastener **1350**. The anti-rotation fastener **1350** may include a head portion **1355**, a shank portion **1357**, a threaded portion **1360**, a tip portion **1362**, a cannulation **1365**, and a rounded edge **1368**.

[0461] The anti-rotation fastener **1350** may be configured to secure the plate **1302** to a vertebra. The anti-rotation fastener **1350** may also be configured to be captively received within the plate **1302** while still being allow to translate along a longitudinal axis of the first slot **1305** and/or the second slot **1315**. The anti-rotation fastener **1350** may be configured so that, when a head portion **1355** engages the first slot **1305** and/or the second slot **1315**, the anti-rotation fastener **1350** is prevented from rotating in relation to the plate **1302**.

[0462] The head portion **1355** may be configured to be received within the first slot **1305** and/or the second slot **1315**. The threaded portion **1360** may be larger than a width of the first slot **1305** and/or the second slot **1315** to prevent the anti-rotation fastener **1350** from disengaging from the plate **1302**. The first threaded feature **1310** and the second threaded feature **1320** may each be configured to threadably engage the threaded portion **1360** to allow the threaded portion **1360** to pass through the first slot **1305** and the second slot **1315** respectively.

[0463] The shank portion **1357** may be smaller than a width of the first slot **1305** and/or the second slot **1315** to allow the anti-rotation fastener **1350** to translate along the length of the first slot **1305** and/or the second slot **1315** respectively. The tip portion **1362** may be configured to ease insertion of the anti-rotation fastener **1350** into a hole in a vertebra. The cannulation **1365** may be configured to slidably receive a tether **60**. The rounded edge **1368** may reduce stress on the tether **60** when the tether **60** exerts a force on the anti-rotation fastener **1350**.

[0464] FIG. **114** is a front view of a fracture plating system **1400** according to an embodiment of the present disclosure spanning an exemplary bone fracture. FIG. **115** is a partial perspective view of the fracture plating system **1400**. FIG. **116** is a partial section view of the fracture plating system **1400**. FIG. **117** is a front section view of the fracture plating system **1400**.

[0465] The fracture plating system **1400** may include similar features as other fracture plating systems previously described within the present disclosure. The fracture plating system **1400** may include a ratchet fastener **1450**, a ratchet cap **1470**, and one of any of the plates previously described, for example, plate **102**.

[0466] The ratchet fastener **1450** may be received in a first slot **105** and/or a second slot **115** of the plate **102**. A ratchet cap **1470** may then be engaged with a ratchet portion **1460** of the ratchet fastener **1450** to hold the ratchet fastener **1450** captively received within the plate **102** while still allowing translation of the ratchet fastener **1450** along a longitudinal axis of a first slot **105** and/or a second slot **115**.

[0467] FIG. **118**A is a bottom perspective view of a ratchet cap **1470** of the fracture plating system **1400** according to an embodiment of the present disclosure. FIG. **118**B is front perspective view of the ratchet cap **1470**, FIG. **118**C is a front view of the ratchet cap **1470**, and FIG. **118**D is a side view of the ratchet cap **1470**. The ratchet portion **1472** may be configured to engage the ratchet portion **1460** of the ratchet fastener **1450** to secure the plate **102** to a portion of a rib.

[0468] The ratchet cap **1470** may include a ratchet portion **1472**, a body **1474**, a body diameter **1476**, a head diameter **1478**, a body length **1480**, and one or more slots **1485**. The ratchet portion **1472** may be configured to engage the ratchet portion **1460** of the ratchet fastener **1450**. The body diameter **1476** may be configured to be received in a hole drilled into a portion of bone. The body length **1480** may be configured to be less than a thickness of a portion of the bone.

[0469] The head diameter **1478** may be configured to be larger than a hole drilled into a portion of bone. The ratchet portion **1472** may include a plurality of grooves. Each of the plurality of grooves may have a leading angle **1481** and a trailing angle **1482**. The leading angle **1481** may be within a range of 15° to 45°. More specifically, the leading angle **1481** may be within a range of 20° to 40°. More specifically, the leading angle **1481** may be 30°. The trailing angle **1482** may be generally perpendicular to a long axis of the body **1474**. The one or more slots **1485** may be configured to allow the body diameter **1476** to temporarily increase as the ratchet portion **1472** of the ratchet cap **1470** engages and the ratchet portion **1460** of the ratchet fastener **1450**.

[0470] FIG. **119**A is a perspective view of a ratchet fastener **1450** of the fracture plating system **1400** according to an embodiment of the present disclosure. FIG. **119**B is a front view of the ratchet fastener **1450** and FIG. **119**C is a side view of the ratchet fastener **1450**. The ratchet fastener **1450** may include a head portion **1455**, a shank portion **1457**, a ratchet portion **1460**, a tip portion **1462**, a cannulation **1465**, and a rounded edge **1468**. The head portion **1455** may be configured to be received within the first slot **105** and/or the second slot **115**. The ratchet portion **1460** may be larger than a width of the first slot **105** and/or the second slot **115** to prevent the ratchet fastener **1450** from disengaging from the plate **102**.

[0471] The shank portion **1457** may be smaller than a width of the first slot **105** and/or the second slot **115** to allow the ratchet fastener **1450** to translate along the length of the first slot **105** and/or the second slot **115**. The tip portion **1462** may be configured to ease insertion of the ratchet fastener **1450** into a hole in a bone portion. The cannulation **1465** may be configured to slidably receive a tether **60**. The ratchet portion **1460** may be configured to receive a ratchet cap **1470**. The rounded edge **1468** may reduce stress on the tether **60** when the tether **60** exerts a force on the ratchet fastener **1450**.

[0472] The ratchet portion **1460** may include a plurality of grooves. Each of the plurality of grooves may have a leading angle **1458** and a trailing angle **1459**. The leading angle **1458** may be within a range of 15° to 45°. More specifically, the leading angle **1458** may be within a range of 20° to 40°. More specifically, the leading angle **1458** may be 30°. The trailing angle **1459** may be generally perpendicular to a long axis of the shank portion **1457**.

[0473] The ratchet fastener **1450** may be configured within a range of lengths from 5 mm to 30 mm and within a range of diameters from 3 mm to 8 mm. The ratchet fastener **1450** may be one of a set of differently-sized implants, each having a different length and/or diameter. The ratchet cap **1470** may be one of a set of differently-sized implants, each having a different diameter configured to engage each of the differently sized ratchet fasteners **1450**.

[0474] The ratchet fastener **1450** may be configured to receive the ratchet cap **1470** so that the ratchet portion **1460** of the ratchet fastener **1450** engages the ratchet portion **1472** of the ratchet cap **1470**. More specifically, the leading angle **1481** of the ratchet cap **1470** and the leading angle **1458** of the ratchet fastener **1450** may be configured to allow the ratchet cap **1470** to advance along a length of the ratchet fastener **1450** from the tip portion **1462** towards the head portion **1455**.

[0475] Moreover, the trailing angle **1482** of the ratchet cap **1470** and the trailing angle of the ratchet fastener **1450** may be configured to prevent translation of the ratchet cap **1470** along a length of the ratchet fastener **1450** from the head portion **1455** towards the tip portion **1462**.

[0476] The ratchet cap **1470** may create compression at the interface similar to the locking nut **1370**, but the advantage may be that it the ratchet cap **1470** may not require a drive feature to install the ratchet cap **1470** and secure the construct. The drive feature on the previous concepts may require the cap to have some thickness to engage with. The ratchet cap **1470** may not require a non-circular drive feature to apply torque. Thus, the ratchet cap **1470** may be much thinner than the locking nut **1370**. A thinner cap may result in a lower profile construct on the bone and thus may reduce irritation of the surrounding soft tissues.

[0477] FIG. **120**A is a perspective view of a fracture repair system **1500** in an undeformed configuration **1520** according to an embodiment of the present disclosure. FIG. **120**B is a front view of the fracture repair system **1500** in an undeformed configuration **1520**. FIG. **120**C is a side view of the fracture repair system **1500** in an undeformed configuration **1520**. The fracture repair system **1500** may be configured as a staple.

[0478] The fracture repair system **1500** may be configured to span a fracture in a portion of a bone. The fracture repair system **1500** may further be configured to provide compression of the fracture after implantation.

[0479] The fracture repair system **1500** may include a bridge portion **1550**, a first leg **1555** having a first tip **1557**, a second leg **1560** having a second tip **1562**, a plurality of retaining features **1570**, and a bridge apex **1580**. The bridge portion **1550** may connect the first leg **1555** and the second leg **1560**. Each of the first leg **1555** and the second leg **1560** may include a plurality of retaining features **1570**. The plurality of retaining features **1570** may be configured to inhibit subsidence or back-out of the fracture repair system **1500** from a bone portion after implantation.

[0480] The first tip **1557** and the second tip **1562** may each be configured with a sharp point configured to penetrate a portion of bone. The fracture repair system **1500** may be configured to be embedded directly into a portion of bone. Additionally, or alternatively, the fracture repair system **1500** may be configured to be embedded into one or more apertures in a portion of bone and/or a portion of cartilage.

[0481] The fracture repair system **1500** may be fabricated from NITINOL, titanium, titanium alloy, stainless steel, cobalt-chrome, PEEK, PEAK, UHMWPE, a resorbable polymer, or any other biocompatible material with sufficient tensile strength and shape memory properties.

[0482] The fracture repair system **1500** may include an undeformed configuration **1520** and an expanded configuration **1540**. FIG. **121** is a front view of the fracture repair system **1500** in an undeformed configuration **1520**. The FIG. **122** is a front view of the fracture repair system **1500** in an expanded configuration **1540** according to an embodiment of the present disclosure.

[0483] In the undeformed configuration **1520** the bridge portion **1550** may include a bridge apex **1580**. The bridge apex **1580** may be centrally located along the bridge portion **1550**. In the undeformed configuration **1520** the fracture repair system **1500** may include a first leg spacing **1590**. In the expanded configuration **1540** the fracture repair system **1500** may include a second leg spacing **1592**. The first leg spacing **1590** may be less than the second leg spacing **1592**.

[0484] The fracture repair system **1500** may be configured so that an external force is required to transition from the undeformed configuration **1520** to the expanded configuration **1540**. Additionally, when the external force is removed, the fracture repair system **1500** will return from the expanded configuration **1540** to the undeformed configuration **1520** with no other externally applied forces.

[0485] An instrument/tool (not shown) may be used to elastically deform the fracture repair system **1500** from the undeformed configuration **1520** to the expanded configuration **1540**. Once inserted into the tissue (bone and/or cartilage) the force of the instrument may be removed and the fracture repair system **1500** may return to the undeformed configuration **1520**. The fracture repair system **1500** may be inserted to bridge a rib fracture in the expanded configuration **1540**, and then may be used to compress the fractured bones back together as it moves back towards the undeformed configuration **1520**.

[0486] FIG. **123** is a front section view of an exemplary rib cage with the fracture repair system **1500** spanning exemplary fractures. FIG. **124** is a front section view of an exemplary rib cage with the fracture repair system **1500** compressing exemplary fractures. FIG. **125** is a front section view of an exemplary rib cage with the fracture repair system **1500** compressing exemplary fractures. The fracture repair system **1500** may be utilized to bridge fractures in ribs, in the costal cartilage, between sternum-to-costal cartilage, rib-to-costal cartilage, bone-to-cartilage, bone-to-bone, and/or cartilage-to-cartilage. One or more of the fracture repair systems **1500** may be applied to a single fracture and may provide additional stabilization and/or additional compression.

[0487] A fracture repair system **1600** may include a balloon **1610**, cement, and an application instrument **1620**. Human rib bones are generally hollow, similar to long bones, and may have an intramedullary canal. The balloon **1610** may be configured for repairing a fracture of any bone having an intramedullary canal. The balloon **1610** may be configured to be inserted into an intramedullary canal and span a fracture from an inside of a portion of bone. The balloon **1610** may then be filled with a bone cement **1650** know in the orthopedic arts. After a period of time and/or application of a hardening agent the bone cement **1650** may harden. The hardened bone cement **1650** and balloon may stabilize the fracture.

[0488] FIG. **126**A is a perspective view of an application instrument **1620** of a fracture repair system **1600** according to an embodiment of the present disclosure. FIG. **126**B is a side view of the application instrument **1620** and FIG. **126**C is a front view of the application instrument **1620**. The application instrument may include a funnel **1622**, a shank **1625**, an aperture **1630**, and an end portion **1635**.

[0489] The shank **1625** may be hollow and may connect the funnel **1622** with the aperture **1630**. The funnel **1622** may be configured to ease insertion of the balloon **1610** and/or bone cement **1650**. The end portion **1635** may be configured to prevent discharge from the shank **1625** through the end portion **1635**. The application instrument **1620** may be configured so that the balloon **1610** and/or the bone cement **1650** may be introduced into the funnel, travel through the interior of the shank **1625** and exit through the aperture **1630**. The aperture **1630** may be configured so that the balloon **1610** and/or the bone cement **1650** exit at a generally right angle with respect to an axis of the shank **1625**.

[0490] FIG. **127** is a balloon **1610** of the fracture repair system **1600** according to an embodiment of the present disclosure. The balloon **1610** may be configured to be deployed through the application instrument **1620** into an intramedullary canal of a bone portion. The balloon **1610** may include an opening **1612** and a body **1614**. The opening **1612** may be sized generally the same as, or slightly larger than, the aperture **1630** to facilitate introduction of bone cement **1650** through the aperture into the interior of the balloon **1610**. The circumference of the balloon **1610** may be generally equal to or larger than the circumference of the intramedullary canal of the fractured bone portion.

[0491] The body **1614** may be configured to receive bone cement **1650** through the opening **1612** and expand upon introduction of the bone cement **1650**. The balloon **1610** may be configured within a range of outer profiles from 8 mm to 25 mm and within a range of lengths from 5 mm to 500 mm. The balloon **1610** may be one of a set of differently-sized implants, each having a different outer profile and/or length. The balloon **1610** may be impregnated with barium sulfate or similar contrast agent. The outer profile may be generally equal to an outside perimeter of the balloon **1610**. In one embodiment, the balloon **1610** may be circular in cross-section and the outer profile may equal the circumference of the cross-section.

[0492] A method for stabilizing one or more fractures of a portion of bone may include using a balloon **1610** and bone cement **1650**. A method for stabilizing one or more fractures of a portion of bone may include the following steps:

[0493] FIG. **128** is a partial perspective view of an exemplary rib cage **20** with an exemplary bone first fracture **21**.

[0494] Step 1 may include finding a fracture **21** using ultrasound or other radiographic means.

[0495] FIG. **129** is a partial perspective view of the exemplary rib cage **20** illustrating a step in a method of deploying a fracture repair system **1600** according to an embodiment of the present disclosure.

[0496] Step 2 may include drilling a first hole **15** through the proximal cortex of the bone portion adjacent to the first fracture **21**.

[0497] FIG. **130** is a partial perspective view of the exemplary rib cage **20** illustrating a step in a method of deploying a fracture repair system **1600** according to an embodiment of the present disclosure.

[0498] Step 3 may include inserting the application instrument **1620** into the hole **15** and orienting the aperture **1630** towards the first fracture **21**.

[0499] FIG. **131** is a partial perspective view of the exemplary rib cage **20** illustrating a step in a method of deploying a fracture repair system **1600** according to an embodiment of the present disclosure.

[0500] Step 4 may include pressurizing the balloon **1610** with air, other gas, and/or saline through the application instrument **1620** into the intramedullary canal towards the first fracture **21**. Verifying that the balloon **1610** has bridged the fracture and entered the opposite portion of the bone using fluoroscopy.

[0501] FIG. **132** is a partial perspective view of the exemplary rib cage **20** illustrating a step in a method of deploying a fracture repair system **1600** according to an embodiment of the present disclosure.

[0502] Step 5 may include pressurizing the balloon **1610** with bone cement **1650** introduced through the application instrument **1620**.

[0503] FIG. **133** is a partial perspective view of the exemplary rib cage **20** illustrating a step in a method of deploying a fracture repair system **1600** according to an embodiment of the present disclosure.

[0504] Step 6 may include continuing to expand the balloon **1610** until the balloon **1610** if fully expanded and spans the first fracture **21**, and using fluoroscopy to verify the position of the balloon **1610** in relation to the first fracture **21**.

[0505] FIG. **134** is a partial perspective view of the exemplary rib cage **20** illustrating a step in a method of deploying a fracture repair system **1600** according to an embodiment of the present disclosure. The balloon **1610** is shown fully expanded within the intramedullary canal and spanning the fracture.

[0506] Step 7 may include removing the application instrument **1620** and allowing the bone cement **1650** to harden.

[0507] FIG. **135** is a partial perspective section view of the exemplary rib cage **20** illustrating a step in a method of deploying a fracture repair system **1600** according to an embodiment of the present disclosure. The section view of the balloon **1610** shows the bone cement **1650** spanning the first fracture **21**.

[0508] FIG. **136** is a partial perspective view of the exemplary rib cage **20** illustrating a step in a method of deploying a fracture repair system **1600** according to an embodiment of the present disclosure. The final construct is shown with the balloon **1610** within the intramedullary canal of a portion of bone and spanning the first fracture **21**.

[0509] A method for stabilizing one or more fractures of a bone may include using a single tether to advance a plate to an interior surface of a bone, using a single tether to advance a first fastener and/or a second fastener to the interior surface of the bone, and/or using a single tether to compress or reduce one or more fractures of a bone. Additionally, or alternatively, the method may include using a second tether to advance a third fastener and/or a fourth fastener to the interior surface of the bone.

[0510] A method for stabilizing one or more fractures of a bone may also include introducing a fracture plating system through a VATS portal. A method for stabilizing one or more fractures of a bone may also include introducing a fracture plating system, with a first fastener captive in a first slot of the plate and/or a second fastener captive in a second slot of the plate, through a VATS portal.

[0511] A method for stabilizing one or more fractures of a bone may also include receiving fasteners through the bone on opposite sides of each of the one or more fractures. Additionally, or alternatively, the method may include receiving two or more fasteners in a slot of the plate.

[0512] A method for stabilizing one or more fractures of a portion of bone may include the following steps:

[0513] FIG. **84** is a perspective view of an exemplary rib cage **20** with an exemplary bone first fracture **21**. FIG. **85** is a partial perspective view of the exemplary rib cage **20**.

[0514] Step 1 may include identifying one or more fractures of a bone.

[0515] FIG. **86** is a partial perspective view of the exemplary rib cage **20** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.

[0516] Step 2 may include drilling a first hole **15** and a second hole **16** wherein the first hole **15** and the second hole **16** are separated by a first fracture **21**.

[0517] FIG. **87** is a partial perspective view of the exemplary rib cage **20** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure. FIG. **88** is a partial perspective view of the exemplary rib cage **20** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.

[0518] Step 3 may include inserting a first end **72** of a first flexible guide tube **70** through the first hole **15**. Inserting a first end **72** of a second flexible guide tube **70** through the second hole **16**. Pulling the first end **72** of the first flexible guide tube **70** and the first end of the second flexible guide tube out **70** through a VATS portal using an endoscopic grasper or similar instrument. Ensuring that a second end **74** of the first flexible guide tube **70** does not pass through the first hole **15** and ensuring that a second end **74** of the second flexible guide tube **70** does not pass through the second hole **16**. The flexible guide tube **70** may be fabricated of silicone rubber or other biocompatible elastomeric material.

[0519] FIG. **89**A is a front view of a set of differently sized plates **102** according to an embodiment of the present disclosure. The fracture plating system **100** may be configured as a kit including a plurality of plates each having a different size and/or configuration, for example different plate length, different plate radius, different plate width, different number of slots, and/or different configuration of slots. The fracture plating system **100** may include a first plate **102***a* having a first plate length **103***a*, and a second plate **102***b* having a second plate length **103***b*, different from the first plate length **103***a*.

[0520] FIG. **89**B is a front view of a set of differently sized fasteners **180** according to an embodiment of the present disclosure. The fracture plating system kit described above may further include a plurality of fasteners **180** each having a different size and/or configuration, for example, different length, different diameter, different thread configuration, and/or different head configuration. The fracture plating system **100** may include a first fastener **180***a* having a first fastener length **181***a*, and a second fastener **180***b* having a second fastener length **181***b*, different than the first fastener length **181***a*.

[0521] FIG. **89**C is a front view of a set of differently sized locking nuts **1170** according to an embodiment of the present disclosure. The fracture plating system kit described above may further include a plurality of locking nuts **1170** each having a different size and/or configuration, for example, different length, different diameter, different thread configuration, and/or different head configuration. The fracture plating system **100** may include a first locking nut **1170***a* having a first

body length **1180***a*, and a second locking nut **1170***b* having a second body length **1180***b*, different than the first body length **1180***a*. The first locking nut **1170***a* and the second locking nut **1170***b* are interchangeably securable to the first fastener **180***a* and the second fastener **180***b*.
[0522] FIG. **89**D is a partial perspective view of an exemplary rib cage **20** showing an exemplary bone first fracture **21**.
[0523] Step 4 may include measuring the rib thickness and other features of the portion of rib near the first fracture **21**. Selecting a plate from a range of available sizes based on patient anatomy and a location of the fracture. Selecting two or more fasteners from a range of available sizes based on patient anatomy and a location of the fracture.
[0524] FIG. **90** is a front view of a fracture plating system illustrating a step in a method of deploying the fracture plating system according to an embodiment of the present disclosure. FIG. **91** is a front view of the fracture plating system of FIG. **90** illustrating a step in a method of deploying the fracture plating system according to an embodiment of the present disclosure. FIG. **92** is a front view of the fracture plating system of FIG. **90** illustrating a step in a method of deploying the fracture plating system according to an embodiment of the present disclosure.
[0525] Step 5 may include assembling the selected fasteners to the selected plate.
[0526] FIG. **93** is a partial perspective view of the exemplary rib cage **20** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.
[0527] Step 6 may include inserting a first tether end **62** through a first fastener and a second tether end **64** through a second fastener.
[0528] FIG. **94** is a partial perspective view of the exemplary rib cage **20** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.
[0529] Step 7 may include inserting a first tether end **62** into a second end **74** of the first flexible guide tube **70** and inserting the second tether end **64** into a second end **74** of the second flexible guide tube **70**.
[0530] FIG. **95** is a partial perspective view of the exemplary rib cage **20** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.
[0531] Step 8 may include feeding the first tether end **62** and the second tether end **64** through the first flexible guide tube **70** and the second flexible guide tube **70**, respectively until the first tether end **62** and the second tether end **64** protrude through the first end **72** of the first flexible guide tube **70** and the first end **72** of the second flexible guide tube **70**.
[0532] FIG. **96** is a partial perspective view of the exemplary rib cage **20** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.
[0533] Step 9 may include removing the first flexible guide tube **70** and the second flexible guide tube **70** from the surgical site by pulling the first end **72** of the first flexible guide tube **70** and the first end **72** of the second flexible guide tube **70**.
[0534] FIG. **97** is a partial perspective view of the exemplary rib cage **20** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.
[0535] Step 10 may include pull the first tether end **62** and the second tether end **64** until the first fastener protrudes through the first hole **15** and the second fastener protrudes through the second hole **16**. Optionally, continuing to pull until the fracture is reduced.
[0536] FIG. **98** is a partial perspective view of the exemplary rib cage **20** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.
[0537] Step 11 may include placing a first washer over the first tether end **62** and then over the tip

portion of the first fastener and placing a second washer over the second tether end **64** and then over the tip portion of the second fastener.

[0538] FIG. **99** is a partial perspective view of the exemplary rib cage **20** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure. FIG. **100** is a partial perspective view of the exemplary rib cage **20** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.

[0539] Step 12 may include placing a first locking nut **30** over the first tether end **62** and threading and tightening the first locking nut onto the threaded portion of the first fastener. Placing a second locking nut **30** over the second tether end **64** and threading and tightening the second locking nut onto the threaded portion of the second fastener.

[0540] FIG. **101** is a partial perspective view of the exemplary rib cage **20** illustrating a step in a method of deploying a fracture plating system according to an embodiment of the present disclosure.

[0541] Step 13 may include removing the single tether **60** by pulling it with an endoscopic grasper, or similar instrument, through the VATS portal.

[0542] A method for stabilizing one or more fractures of a bone may include using a tether to advance a first fastener and/or a second fastener to the interior surface of the bone. The method may further include introducing the first fastener and/or the second fastener through a VATS portal. The method may also include introducing the plate to an exterior surface of the bone. The method may include advancing the fasteners through a hole in the bone so that a head portion contacts the interior surface and a distal end extends from the exterior surface of the bone. The method may further include receiving the distal end within a slot of the plate, and, using the fastener, securing the plate to the exterior surface of the bone.

[0543] A method for stabilizing one or more fractures of a portion of bone may include the following steps: [0544] Step 1 may include identifying one or more fractures of a bone. [0545] Step 2 may include drilling a first hole **15** and a second hole **16** wherein the first hole **15** and the second hole **16** are separated by a first fracture **21**. The step may further include drilling additional holes through the bone on opposite sides of additional fractures, so that each fracture is between two holes in the bone. [0546] Step 3 may include inserting a first end **72** of a first flexible guide tube **70** through the first hole **15**. Inserting a first end **72** of a second flexible guide tube **70** through the second hole **16**. Pulling the first end **72** of the first flexible guide tube **70** and the first end of the second flexible guide tube out **70** through a VATS portal using an endoscopic grasper or similar instrument. Ensuring that a second end **74** of the first flexible guide tube **70** does not pass through the first hole **15** and ensuring that a second end **74** of the second flexible guide tube **70** does not pass through the second hole **16**. The flexible guide tube **70** may be fabricated of silicone rubber or other biocompatible elastomeric material. [0547] Step 4 may include measuring the rib thickness and other features of patient anatomy and a location of the fracture. Selecting two or more fasteners from a range of available sizes based on patient anatomy and a location of the fracture. Selecting a plate from a range of available sizes based on patient anatomy and location/quantity of one or more fractures. [0548] Step 5 may include inserting a first tether end **62** through a first fastener and a second tether end **64** through a second fastener. [0549] Step 6 may include inserting a first tether end **62** into a second end **74** of the first flexible guide tube **70** and inserting the second tether end **64** into a second end **74** of the second flexible guide tube **70**. [0550] Step 7 may include feeding the first tether end **62** and the second tether end **64** through the first flexible guide tube **70** and the second flexible guide tube **70**, respectively until the first tether end **62** and the second tether end **64** protrude through the first end **72** of the first flexible guide tube **70** and the first end **72** of the second flexible guide tube **70**. [0551] Step 8 may include removing the first flexible guide tube **70** and the second flexible guide tube **70** from the surgical site by pulling the first end **72** of the first flexible guide tube **70** and the first end **72** of the second flexible guide tube **70**. [0552] Step 9 may

include pulling the first tether end **62** and the second tether end **64** until the first fastener is received in the first hole **15** and the second fastener is received in the second hole **16**. [0553] Step 10 may optionally include repeating Step 3 through Step 9 to introduce additional fasteners through additional holes in the bone. [0554] Step 11 may include placing the plate on an exterior surface of the bone so that the plate spans the one or more fractures and the fasteners are received within one or more slots of the plate. The plate may be guided to the exterior surface by receiving the first tether end **62** and the second tether end **64** through one or more slots of the plate. The plate may be placed on the exterior surface of the bone through an open incision and/or through a transverse skin tunnel connecting two vertical soft tissue tunnels. [0555] Step 12 may include securing the plate to the exterior surface of the bone by placing a first locking nut **30** over the first tether end **62** and threading and tightening the first locking nut onto the threaded portion of the first fastener and placing a second locking nut **30** over the second tether end **64** and threading and tightening the second locking nut onto the threaded portion of the second fastener. Optionally, additional locking nuts **30** may be used to engage additional fasteners. [0556] Step 13 may include removing the single tether **60** by pulling it with an endoscopic grasper, or similar instrument, through the VATS portal and/or a trans-intercostal incision.
[0557] Those of skill in the art will recognize that this is only one of many potential methods that may be used to stabilize one or more fractures of a bone. In alternative embodiments, different methods may be used to implant the fracture plating system **100** or other fracture plating systems described above. Further, the method set forth in FIG. **84** though FIG. **101** may be used to implant other fracture plating systems besides those specifically disclosed herein.
[0558] Reference throughout this specification to “an embodiment” or “the embodiment” means that a particular feature, structure or characteristic described in connection with that embodiment is included in at least one embodiment. Thus, the quoted phrases, or variations thereof, as recited throughout this specification are not necessarily all referring to the same embodiment.
[0559] Similarly, it should be appreciated that in the above description of embodiments, various features are sometimes grouped together in a single embodiment, Figure, or description thereof for the purpose of streamlining the disclosure. This method of disclosure, however, is not to be interpreted as reflecting an intention that any claim require more features than those expressly recited in that claim. Rather, as the following claims reflect, inventive aspects lie in a combination of fewer than all features of any single foregoing disclosed embodiment. Thus, the claims following this Detailed Description are hereby expressly incorporated into this Detailed Description, with each claim standing on its own as a separate embodiment. This disclosure includes all permutations of the independent claims with their dependent claims.
[0560] The phrases “generally parallel” and “generally perpendicular” refer to structures that are within 30° parallelism or perpendicularity relative to each other, respectively. Recitation in the claims of the term “first” with respect to a feature or element does not necessarily imply the existence of a second or additional such feature or element. Elements recited in means-plus-function format are intended to be construed in accordance with 35 U.S.C. § 112 Para. 6. It will be apparent to those having skill in the art that changes may be made to the details of the above-described embodiments without departing from the underlying principles of the disclosure.
[0561] While specific embodiments and applications of the present disclosure have been illustrated and described, it is to be understood that the disclosure is not limited to the precise configuration and components disclosed herein. Various modifications, changes, and variations which will be apparent to those skilled in the art may be made in the arrangement, operation, and details of the methods and systems of the present disclosure without departing from its spirit and scope.

Claims

1. A fracture plating system configured to stabilize a first fracture of a bone of a patient, the bone comprising an interior surface facing toward an interior body cavity of the patient, and an exterior surface facing away from the interior body cavity, the fracture plating system comprising: a plate comprising one or more slots and configured to span the first fracture; a first fastener configured to be received in one of the one or more slots and secure the plate to the interior surface of the bone; a second fastener configured to be received in one of the one or more slots and secure the plate to the interior surface of the bone; and a first tether configured to apply a force directly to the first fastener and the second fastener to guide the plate, the first fastener, and the second fastener through the interior body cavity to the interior surface of the bone.
2. The fracture plating system of claim 1, wherein the first tether is further configured to facilitate reduction of the first fracture by, with the first fastener and the second fastener secured to the interior surface on opposite sides of the first fracture, urging the first fastener and the second fastener towards each other.
3. The fracture plating system of claim 1, wherein the first tether is further configured to guide the first fastener through a first hole in the bone and to guide the second fastener through a second hole in the bone, wherein the first hole and the second hole are on opposite sides of the first fracture.
4. The fracture plating system of claim 1, wherein: the bone further comprises a second fracture; and the plate is further configured to stabilize the second fracture via attachment of the plate to the bone such that the plate spans the first fracture and the second fracture; and the fracture plating system further comprises: a third fastener configured to be received in one of the one or more slots and secure the plate to the interior surface of the bone; a fourth fastener configured to be received in one of the one or more slots and secure the plate to the interior surface of the bone; and a second tether configured to guide the third fastener and the fourth fastener to the interior surface of the bone.
5. The fracture plating system of claim 4, wherein the second tether is further configured to guide the third fastener to a first slot of the one or more slots and to guide the fourth fastener to a second slot of the one or more slots.
6. The fracture plating system of claim 1, further configured to stabilize three or more fractures of the bone, wherein the plate is further configured to span the three or more fractures and the fracture plating system further comprises multiple tethers each configured to guide two fasteners to the interior surface of the bone and to one of the one or more slots.
7. The fracture plating system of claim 1, wherein: the bone comprises a rib; and the first tether is further configured to draw the plate, the first fastener, and the second fastener through a VATS portal to the interior surface of the bone.
8. A fracture plating system configured to stabilize a first fracture of a bone of a patient, the bone comprising an interior surface facing toward an interior body cavity of the patient, and an exterior surface facing away from the interior body cavity, the fracture plating system comprising: a plate comprising one or more slots and configured to span the first fracture; a first fastener configured to be received in one of the one or more slots and secure the plate to the interior surface of the bone; a second fastener configured to be received in one of the one or more slots and secure the plate to the interior surface of the bone; and a first tether configured to apply a force directly to the first fastener and the second fastener to guide the plate, the first fastener, and the second fastener through the interior body cavity to the interior surface of the bone.
9. The fracture plating system of claim 8, wherein the first tether is further configured to facilitate reduction of the first fracture by, with the first fastener and the second fastener secured to the interior surface on opposite sides of the first fracture, urging the first fastener and the second fastener towards each other.
10. The fracture plating system of claim 8, wherein: the bone comprises a rib; and the first tether is further configured to draw the plate, the first fastener, and the second fastener through a VATS

portal to the interior surface of the bone.

11. The fracture plating system of claim 8, wherein: the bone further comprises a second fracture; and the plate is further configured to stabilize the second fracture via attachment of the plate to the bone such that the plate spans the first fracture and the second fracture; and the fracture plating system further comprises: a third fastener configured to be received in one of the one or more slots and secure the plate to the interior surface of the bone; a fourth fastener configured to be received in one of the one or more slots and secure the plate to the interior surface of the bone; and a second tether configured to guide the third fastener and the fourth fastener to the interior surface of the bone.

12. The fracture plating system of claim 8, wherein the first fastener comprises a first cannulation configured to receive a first end of the first tether, and the second fastener comprises a second cannulation configured to receive a second end of the first tether.

13. The fracture plating system of claim 8, further comprising a first locking nut configured to receive the first fastener and to cooperate with the first fastener to secure the plate to the interior surface of the bone, and a second locking nut configured to receive the second fastener and to cooperate with the second fastener to secure the plate to the interior surface of the bone.

14. A fracture plating system configured to stabilize a first fracture of a bone of a patient, the bone comprising an interior surface facing toward an interior body cavity of the patient, and an exterior surface facing away from the interior body cavity, the fracture plating system comprising: a plate assembly comprising: a plate comprising one or more slots and configured to span the first fracture; a first fastener coupled to one of the one or more slots and configured to secure the plate to the interior surface of the bone; and a second fastener coupled to one of the one or more slots and configured to secure the plate to the interior surface of the bone; and a first tether configured to apply a force directly to the first fastener and the second fastener to guide the plate assembly through the interior body cavity to the interior surface of the bone.

15. The fracture plating system of claim 14, wherein the first tether is further configured to draw the first fastener through a first hole in a first portion of the bone proximate a first side of the first fracture; and draw the second fastener through a second hole in a second portion of the bone proximate a second side of the first fracture.

16. The fracture plating system of claim 14, wherein: the bone further comprises a second fracture; and the plate is further configured to stabilize the second fracture via attachment of the plate to the bone such that the plate spans the first fracture and the second fracture; and the fracture plating system further comprises: a third fastener configured to be received in one of the one or more slots and secure the plate to the interior surface of the bone; a fourth fastener configured to be received in one of the one or more slots and secure the plate to the interior surface of the bone; and a second tether configured to guide the third fastener and the fourth fastener to the interior surface of the bone.

17. The fracture plating system of claim 14, wherein the first tether is further configured to facilitate reduction of the first fracture by, with the first fastener and the second fastener secured to the interior surface on opposite sides of the first fracture, urging the first fastener and the second fastener towards each other.

18. The fracture plating system of claim 14, wherein: the bone comprises a rib; and the first tether is further configured to draw the plate assembly through a VATS portal to the interior surface of the bone.

19. The fracture plating system of claim 14, wherein the first fastener comprises a first cannulation configured to receive a first end of the first tether, and the second fastener comprises a second cannulation configured to receive a second end of the first tether.

20. The fracture plating system of claim 14, further comprising a first locking nut configured to receive the first fastener and to cooperate with the first fastener to secure the plate to the interior surface of the bone, and a second locking nut configured to receive the second fastener and to cooperate with the second fastener to secure the plate to the interior surface of the bone.